7.3.16 The table in Figure 7.3.16 gives the values of two functions f and g at certain points. Calculate $(f\circ g)'(1)$ and $(g\circ f)'(1)$. Is there enough information to calculate $(f\circ g)'(3)$ and/or $(g\circ f)'(3)$? How about $\dfrac{d}{dx}(f^2)(1)$ and $(f\circ f)'(1)$?

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	3	3	2	2
2	4	4	4	0
3	6	1	1	0
4	-1	0	1	1
5	2	5	3	3

Figure 7.5. Values of f and g at several points.

7.3.3 Inverse Functions

Suppose that a function $f : I \to J$ has an inverse. This simply means that there is a function g (called the *inverse* of f) that reverses the mapping: If $f(a) = b$ then $g(b) = a$. We can assume that I and J are intervals. Thus f maps the interval I onto the interval J and the inverse function g then maps J back to I. Not all functions have an inverse, but we are supposing that this one does.

Suppose too that f is differentiable at a point $x_0 \in I$. Then we would expect from geometric considerations that that the inverse function g should be differentiable at the image point $z_0 = f(x_0) \in J$.

This is entirely elementary. The connection between a function f and its inverse g is given by

$$f(g(x)) = x \text{ for all } x \in J$$

or

$$g(f(x)) = x \text{ for all } x \in I.$$

Using the chain rule on the second of these immediately gives

$$g'(f(x))f'(x) = 1$$

and hence we have the connection

$$g'(f(x)) = \frac{1}{f'(x)},$$

which a geometrical argument could also have found.

Example 7.12: Suppose that the exponential function e^x has been developed and that we have proved that it is differentiable for all values of x and we have the usual formula $\frac{d}{dx}e^x = e^x$. Then, provided we can be sure there is an inverse, a formula for the derivative of that inverse can be found. Let $L(x)$ be the inverse function of $f(x) = e^x$. Then, since we know that $f'(x) = f(x)$

$$L'(f(x)) = \frac{1}{f'(x)} = \frac{1}{f(x)}$$

or, replacing $f(x)$ by another letter, say z, we have

$$L'(z) = \frac{1}{z}.$$

This must be valid for every value z in the domain of L, that is, for every value in the range of f. You should recognize the derivative of the function $\ln z$ here. Even so, we would still need to justify the existence of the inverse function before we could properly claim to have proved this formula. ◀

We would like a better way to handle inverse functions than presented here. Our observations here allow us to compute the derivative of an inverse but do not assure us that an inverse will exist. For a theorem that allows us merely to look at the derivative and determine that an inverse exists and has a derivative, see Theorem 7.32.

Exercises

7.3.17 Find a formula for the derivative of the function $\sin^{-1} x$ assuming that the usual formula for

$$\frac{d}{dx}\sin x = \cos x$$

has been found.

SEE NOTE 175

7.3.18 Find a formula for the derivative of the function $\tan^{-1} x$ assuming that the usual formula for $\frac{d}{dx}\tan x = \sec^2 x$ has been found.

7.3.19 Give a geometric interpretation of the relationship between the slope of the tangent at a point (x_0, y_0) on the graph of $y = f(x)$ and the slope of the tangent at the point (y_0, x_0) on the graph of $y = g(x)$ where g is the inverse of f.

SEE NOTE 176

7.3.20 What facts about the function $f(x) = e^x$ would need to be established in order to claim that there is indeed an inverse function? What is the domain and range of that inverse function?

7.3.4 The Power Rule

The power rule is the formula

$$\frac{d}{dx}x^p = px^{p-1}$$

which is the basis for many calculus problems. We have already shown (in Exercise 7.3.5) that

$$\frac{d}{dx}x^n = nx^{n-1}$$

for $n = 1, 2, 3, \ldots$ and for every value of x.

This is easy enough to extend to negative integers. Just interpret for $n = 1, 2, 3, \ldots$ and for every value of $x \neq 0$,

$$\frac{d}{dx}x^{-n} = \frac{d}{dx}\frac{1}{x^n}$$

and, using the quotient rule, we find that again the power rule formula is valid for $p = -1, -2, -3, \ldots$ and any value of x other than 0.

The formula also works for $p = 0$ since we interpret x^0 as the constant 1 (although for $x = 0$ we prefer not to make any claims). Is the formula indeed valid for every value of p, not just for integer values?

Example 7.13: We can verify the power rule formula for $p = 1/2$; that is, we prove that

$$\frac{d}{dx}\sqrt{x} = \frac{d}{dx}x^{1/2} = \frac{1}{2}x^{1/2-1} = \frac{1}{2\sqrt{x}}.$$

First we must insist that $x > 0$ otherwise $\sqrt{x}$ and the fraction in our formula would not be defined. Now interpret $\sqrt{x}$ as the inverse of the square function $f(x) = x^2$. Specifically let $f(x) = x^2$ for $x > 0$ and $g(x) = \sqrt{x}$ for $x > 0$ and note that $f(g(x)) = g(f(x)) = x$. Thus

$$\frac{d}{dx}f(g(x)) = \frac{d}{dx}x = 1$$

and so, since $f'(x) = 2x$ and $f'(g(x))g'(x) = 1$ we obtain $2\sqrt{x}g'(x) = 1$ and finally that

$$g'(x) = \frac{1}{2\sqrt{x}}$$

as required if the power rule formula is valid. ◀

Is the power rule

$$\frac{d}{dx}x^p = px^{p-1}$$

valid for all rational values of p? We can handle the case $p = m/n$ for integer m and n by essentially the same methods. We state this as a theorem whose proof is left as an exercise. For irrational p there is also a discussion in the exercises.

Theorem 7.14: *Let $f(x) = x^{\frac{m}{n}}$ for $x > 0$ and integers m, n. Then*

$$f'(x) = \frac{m}{n}x^{\frac{m}{n}-1}.$$

Example 7.15: Every polynomial is differentiable on $\mathbb{R}$ and its derivative can be calculated via term by term differentiation; that is,

$$\frac{d}{dx}(a_0 + a_1x + a_2x^2 + \cdots + a_nx^n) = a_1 + 2a_2x + \cdots + na_nx^{n-1}.$$

This follows from the power rule formula and the rule for sums. Note that the derivative of a polynomial is again a polynomial. ◀

Example 7.16: A rational function is a function $R(x)$ that can be expressed as the quotient of two polynomials,

$$R(x) = \frac{p(x)}{q(x)}.$$

This would be defined at every point at which the denominator $q(x)$ is not equal to zero. Every rational function is differentiable except at those points at which the denominator vanishes. This follows from the previous example, which showed how to differentiate a polynomial, and from the quotient rule. Thus

$$\frac{d}{dx}\left(\frac{p(x)}{q(x)}\right) = \frac{p'(x)q(x) - p(x)q'(x)}{q^2(x)}.$$

Notice that the derivative is another rational function with the same domain since both numerator and denominator are again polynomials. ◀

Exercises

7.3.21 Prove Theorem 7.14.
SEE NOTE 177

7.3.22 Show that the power formula is available for all values of p once the formula $\frac{d}{dx}e^x = e^x$ is known.
SEE NOTE 178

7.3.23 Let

$$p(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n.$$

Compute the sequence of values $p(0)$, $p'(0)$, $p''(0)$, $p'''(0)$, ...
SEE NOTE 179

7.3.24 Determine the coefficients of the polynomial

$$p(x) = (1+x)^n = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$$

by using the formulas that you obtained in Exercise 7.3.23.
SEE NOTE 180

7.4 Continuity of the Derivative?

We have already observed (Theorem 7.6) that if a function f is differentiable on an interval I, then f is also continuous on I. This statement should not be confused with the (incorrect) statement that the derivative, f', is continuous.

Example 7.17: Consider the function f defined on $\mathbb{R}$ by

$$f(x) = \begin{cases} x^2 \sin x^{-1}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0. \end{cases}$$

Since $|\sin x^{-1}| \leq 1$ for all $x \neq 0$, $|f(x)| \leq x^2$ for all $x \in \mathbb{R}$. We can now conclude (e.g., from Example 7.5) that $f'(0) = 0$. For $x \neq 0$, we can calculate, as in elementary calculus, that

$$f'(x) = -\cos x^{-1} + 2x \sin x^{-1}.$$

This function f' is continuous at every point $x_0 \neq 0$. At $x_0 = 0$ it is discontinuous. To see this we need only consider an appropriate sequence $x_n \to 0$ and see what happens

to $f'(x_n)$. For example, try the sequence

$$x_n = \frac{1}{\pi n}.$$

Since

$$\cos\left(\frac{1}{x_n}\right) = \cos(\pi n)$$

and these numbers are alternately $+1$ and -1 it is clear that $f'(x_n)$ cannot converge. Consequently, f' is discontinuous at 0. ◀

Observe that the function f provides an example of a function that is differentiable on all of $\mathbb{R}$, yet f' is discontinuous at a point. It is possible to modify this function to obtain a differentiable function g whose derivative g' is discontinuous at infinitely many points, and even at all the points of the Cantor set (see Exercise 7.4.2).

You might wonder, then, if anything positive could be said about the properties of a derivative f'. It is possible for the derivative of a differentiable function to be discontinuous on a dense set[1]: An example is given later in Section 9.7. We will also show, in Section 7.9, that the function f', while perhaps discontinuous, nonetheless shares one significant property of continuous functions: It has the intermediate value property (Darboux property).

Exercises

7.4.1 Give a simple example of a function f differentiable in a deleted neighborhood of x_0 such that $\lim_{x\to x_0} f'(x)$ does not exist.

7.4.2 ✂ Let P be a Cantor subset of $[0,1]$ (i.e., P is a nonempty, nowhere dense perfect subset of $[0,1]$) and let $\{(a_n,b_n)\}$ be the sequence of intervals complementary to P in $(0,1)$. (See Section 6.5.1.)

(a) On each interval $[a_n,b_n]$ construct a differentiable function such that

$$f_n(a_n) = f_n(b_n) = (f_n')_+(a_n) = (f_n')_-(b_n) = 0,$$
$$\limsup_{x\to a_n^+} f'(x) = \limsup_{x\to b_n^-} f_n'(x) = 1,$$
$$\liminf_{x\to a_n^+} f'(x) = \liminf_{x\to b_n^-} f_n'(x) = -1,$$

and $|f_n(x)| \le (x-a_n)^2(x-b_n)^2$ and $|f_n'(x)|$ is bounded by 1 in each interval $[a_n,b_n]$.

(b) Let g be defined on $[0,1]$ by

$$g(x) = \begin{cases} f_n(x), & \text{if } x \in (a_n,b_n), n = 1,2,\ldots \\ 0, & \text{if } x \in P. \end{cases}$$

Sketch a picture of the graph of g.

(c) Prove that g is differentiable on $[0,1]$.

(d) Prove that $g'(x) = 0$ for each $x \in P$.

(e) Prove that g' is discontinuous at every point of P.

[1] It is not possible for a derivative to be discontinuous at every point. See Corollary 9.40.

7.5 Local Extrema

We have seen in Section 5.7 that a continuous function defined on a closed interval $[a,b]$ achieves an absolute maximum value and an absolute minimum value on the interval. These are called *extreme values* or *extrema*. There must be points where the maximum and minimum are attained, but how do we go about finding such points? One way is to find all points that may not be themselves extrema, but are *local* extreme points. A function defined on an interval I is said to have a *local maximum* at x_0 in the interior of I, if there exists $\delta > 0$ such that $[x_0-\delta, x_0+\delta] \subset I$ and $f(x) \le f(x_0)$ for all x in the smaller interval. A local minimum is similarly defined.

A familiar process studied in elementary calculus is sometimes useful for locating these extrema when the function is differentiable on (a,b): We look for critical points (i.e., points where the derivative is zero). We begin with the theorem that forms the basis for this process.

Theorem 7.18: *Let f be defined on an interval I. If f has a local extremum at a point x_0 in the interior of I and f is differentiable at x_0, then $f'(x_0) = 0$.*

Proof. Suppose f has a local maximum at x_0 in the interior of I, the proof for a local minimum being similar. Then there exists $\delta > 0$ such that

$$[x_0-\delta, x_0+\delta] \subset I \text{ and } f(x) \le f(x_0)$$

for all $x \in [x_0-\delta, x_0+\delta]$. Thus

$$\frac{f(x)-f(x_0)}{x-x_0} \le 0 \text{ for } x \in (x_0, x_0+\delta) \tag{8}$$

and

$$\frac{f(x)-f(x_0)}{x-x_0} \ge 0 \text{ for } x \in (x_0-\delta, x_0). \tag{9}$$

If $f'(x_0)$ exists, then

$$f'(x_0) = \lim_{x\to x_0+} \frac{f(x)-f(x_0)}{x-x_0} = \lim_{x\to x_0-} \frac{f(x)-f(x_0)}{x-x_0}. \tag{10}$$

By (8), the first of these limits is at most zero; by (9), the second is at least zero. By (10), these limits are equal and are therefore equal to zero. ■

It follows from Theorem 7.18 that a function f that is continuous on $[a,b]$ must achieve its maximum at one (or more) of these types of points:

1. Points $x_0 \in (a,b)$ at which $f'(x_0) = 0$
2. Points $x_0 \in (a,b)$ at which f is not differentiable
3. The points a or b

We leave it to you to provide simple examples of each of these possibilities.

The usual process for locating extrema in elementary calculus thus involves locating points at which f has a zero derivative and comparing the values of f at those points

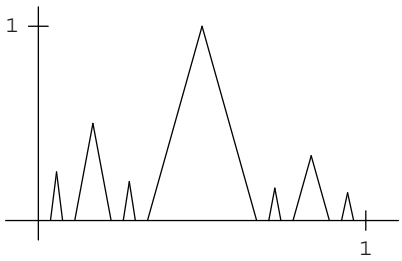


Figure 7.6. Part of the graph of the function in Exercise 7.5.3.

and the points of nondifferentiability (if any) and at the endpoints a and b. In the setting of elementary calculus the situation is usually relatively simple: The function is differentiable, the set on which $f'(x) = 0$ is finite (or contains an interval), and the equation $f'(x) = 0$ is easily solved. Much more complicated situations can occur, of course. The following exercises provide some examples and theorems that indicate just how complicated the set of extrema can be.

Exercises

7.5.1 Give an example of a differentiable function on $\mathbb{R}$ for which $f'(0) = 0$ but 0 is not a local maximum or minimum of f.

7.5.2 Let

$$f(x) = \begin{cases} x^4(2+\sin x^{-1}), & \text{if } x \neq 0 \\ 0, & \text{if } x = 0. \end{cases}$$

(a) Prove that f is differentiable on $\mathbb{R}$.

(b) Prove that f has an absolute minimum at $x = 0$.

(c) Prove that f' takes on both positive and negative values in every neighborhood of 0.

7.5.3 ✂ Let K be the Cantor set and let $\{(a_k, b_k)\}$ be the sequence of intervals complementary to K in $[0,1]$. For each k, let $c_k = (a_k + b_k)/2$. Define f on $[0,1]$ to be zero on K, $1/k$ at c_k, linear and continuous on each of the intervals. (See Figure 7.6.)

(a) Write equations that represent f on the intervals $[a_k, c_k]$ and $[c_k, b_k]$.

(b) Show that f is continuous on $[0,1]$.

(c) Verify that f has minimum zero, achieved at each $x \in K$.

(d) Verify that f has a local maximum at each of the points c_k.

(e) Modify f to a differentiable function with the same set of extrema.

7.5.4 Find all local extrema of the Dirichlet function (see Section 5.2.6) defined on $[0,1]$ by

$$f(x) = \begin{cases} 0, & \text{if } x \text{ is irrational or } x = 0 \\ 1/q, & \text{if } x = p/q,\ p,q \in \mathbb{N},\ p/q \text{ in lowest terms.} \end{cases}$$

7.5.5 Show that the functions in Exercises 7.5.3 and 7.5.4 have infinitely many maxima, all of them strict. Show that the sets of points at which these functions have a strict maximum is countable.

7.5.6 Prove that if $f:\mathbb{R}\to\mathbb{R}$, then $\{x : f \text{ achieves a strict maximum at } x\}$ is countable.
SEE NOTE 181

7.5.7 Let $f:\mathbb{R}\to\mathbb{R}$ have the following property: For each $x\in\mathbb{R}$, f achieves a local maximum (not necessarily strict) at x.

(a) Give an example of such an f whose range is infinite.

(b) Prove that for every such f, the range is countable.

SEE NOTE 182

7.5.8 There are continuous functions $f:\mathbb{R}\to\mathbb{R}$, even differentiable functions, that are nowhere monotonic. This means that there is *no interval* on which the function is increasing, decreasing, or constant. For such functions, the set of maxima as well as the set of minima is dense in $\mathbb{R}$. Construction of such functions is given later in Section 13.14.2. Show that such a function f maps its set of extrema onto a dense subset of the range of f.

7.6 Mean Value Theorem

There is a close connection between the values of a function and the values of its derivative. In one direction this is trivial since the derivative is defined in terms of the values of the function. The other direction is more subtle. How does information about the derivative provide us with information about the function? One of the keys to providing that information is the mean value theorem.

Suppose f is continuous on an interval $[a,b]$ and is differentiable on (a,b). Consider a point x in (a,b). For $y\in(a,b)$, $y\neq x$, the difference quotient

$$\frac{f(y)-f(x)}{y-x}$$

represents the slope of the chord determined by the points $(x,f(x))$ and $(y,f(y))$. This slope may or may not be a good approximation to $f'(x)$. If y is sufficiently near x, the approximation will be good; otherwise it may not be. The mean value theorem asserts that somewhere in the interval determined by x and y there will be a point at which the derivative is exactly the slope of the given chord. It is the existence of such a point that provides a connection between the values of the function [in this case the value $(f(y)-f(x))/(y-x)$] and the value of the derivative (in this case the value at *some* point between x and y).

7.6.1 Rolle's Theorem

We begin with a preliminary theorem that provides a special case of the mean value theorem. This derives its name from Michel Rolle (1652–1719) who has little claim to fame other than this. Indeed Rolle's name was only attached to this theorem because he had published it in a book in 1691; the method itself he did not discover. Perhaps his greatest real contribution is the invention of the notation $\sqrt[n]{x}$ for the nth root of x.

Theorem 7.19 (Rolle's Theorem) *Let f be continuous on $[a,b]$ and differentiable on (a,b). If $f(a)=f(b)$ then there exists $c\in(a,b)$ such that $f'(c)=0$.*

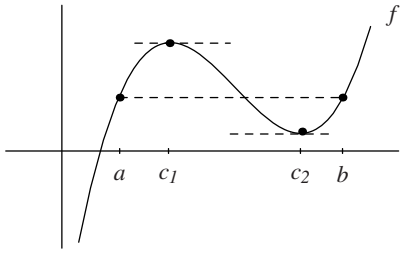


Figure 7.7. Rolle's theorem [note that $f(a) = f(b)$].

Proof. If f is constant on $[a,b]$, then $f'(x) = 0$ for all $x \in (a,b)$, so c can be taken to be any point of (a,b).

Suppose then that f is not constant. Because f is continuous on the compact interval $[a,b]$, f achieves a maximum value M and a minimum value m on $[a,b]$ (Theorem 5.50). Because f is not constant, one of the values M or m is different from $f(a)$ and $f(b)$, say $M > f(a)$. Choose $c \in (a,b)$ such that $f(c) = M$. Since $M > f(a) = f(b)$, $c \neq a$ and $c \neq b$, so $c \in (a,b)$. By Theorem 7.18, $f'(c) = 0$. ■

Observe that Rolle's theorem asserts that under our hypotheses, there is a point at which the tangent to the graph of the function is horizontal, and therefore has the same slope as the chord determined by the points $(a, f(a))$ and $(b, f(b))$. (See Figure 7.7.)

There may, of course, be many such points; Rolle's theorem just guarantees the existence of at least one such point. Observe also that we did not require that f be differentiable at the endpoints a and b. The theorem applies to such functions as $f(x) = x \sin x^{-1}$, $f(0) = 0$, on the interval $[0, 1/\pi]$. This function is not differentiable at zero, but it does have an infinite number of points between 0 and $1/\pi$ where the derivative is zero.

Exercises

7.6.1 Apply Rolle's theorem to the function $f(x) = \sqrt{1-x^2}$ on $[-1,1]$. Observe that f fails to be differentiable at the endpoints of the interval.

7.6.2 Use Rolle's theorem to explain why the cubic equation

$$x^3 + \alpha x^2 + \beta = 0$$

cannot have more than one solution whenever $\alpha > 0$.

7.6.3 If the nth-degree equation

$$p(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n = 0$$

has n distinct real roots, then how many distinct real roots does the $(n-1)$st degree equation $p'(x) = 0$ have?

SEE NOTE 183

7.6.4 Suppose that $f'(x) > c > 0$ for all $x \in [0,\infty)$. Show that $\lim_{x\to\infty} f(x) = \infty$.

7.6.5 Suppose that $f : \mathbb{R} \to \mathbb{R}$ and both f' and f'' exist everywhere. Show that if f has three zeros, then there must be some point ξ so that $f''(\xi) = 0$.

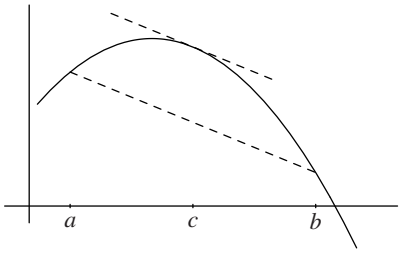


Figure 7.8. Mean value theorem [$f'(c)$ is slope of the chord].

SEE NOTE 184

7.6.6 Let f be continuous on an interval $[a,b]$ and differentiable on (a,b) with a derivative that never is zero. Show that f maps $[a,b]$ one-to-one onto some other interval.

SEE NOTE 185

7.6.7 Let f be continuous on an interval $[a,b]$ and twice differentiable on (a,b) with a second derivative that never is zero. Show that f maps $[a,b]$ two-one onto some other interval; that is, there are at most two points in $[a,b]$ mapping into any one value in the range of f.

SEE NOTE 186

7.6.2 Mean Value Theorem

If we drop the requirement in Rolle's theorem that $f(a) = f(b)$, we now obtain the result that there is a $c \in (a,b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Geometrically, this states that there exists a point $c \in (a,b)$ for which the tangent to the graph of the function at $(c, f(c))$ is parallel to the chord determined by the points $(a, f(a))$ and $(b, f(b))$. (See Figure 7.8.)

This is the mean value theorem, also known as the law of the mean or the first mean value theorem (because there are other mean value theorems).

Theorem 7.20 (Mean Value Theorem) *Suppose that f is a continuous function on the closed interval $[a,b]$ and differentiable on (a,b). Then there exists $c \in (a,b)$ such that*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Proof. We prove this theorem by subtracting from f a function whose graph is the straight line determined by the chord in question and then applying Rolle's theorem. Let

$$L(x) = f(a) + \frac{f(b) - f(a)}{b - a}(x - a).$$

We see that $L(a) = f(a)$ and $L(b) = f(b)$. Now let

$$g(x) = f(x) - L(x). \tag{11}$$

Then g is continuous on $[a,b]$, differentiable on (a,b), and satisfies the condition $g(a) = g(b) = 0$.

By Rolle's theorem, there exists $c \in (a,b)$ such that $g'(c) = 0$. Differentiating (11), we see that $f'(c) = L'(c)$. But

$$L'(c) = \frac{f(b) - f(a)}{b - a},$$

so

$$f'(c) = \frac{f(b) - f(a)}{b - a},$$

as was to be proved. ■

Rolle's theorem and the mean value theorem were easy to prove. The proofs relied on the geometric content of the theorems. We suggest that you take the time to understand the geometric interpretation of these theorems.

Exercises

7.6.8 A function f is said to satisfy a *Lipschitz condition* on an interval $[a,b]$ if

$$|f(x) - f(y)| \leq M|x - y|$$

for all x, y in the interval. Show that if f is assumed to be continuous on $[a,b]$ and differentiable on (a,b) then this condition is equivalent to the derivative f' being bounded on (a,b).

See Note 187

7.6.9 Suppose f satisfies the hypotheses of the mean value theorem on $[a,b]$. Let S be the set of all slopes of chords determined by pairs of points on the graph of f and let

$$D = \{f'(x) : x \in (a,b)\}.$$

(a) Prove that $S \subset D$.

(b) Give an example to show that D can contain numbers not in S.

See Note 188

7.6.10 Interpreting the slope of a chord as an average rate of change and the derivative as an instantaneous rate of change, what does the mean value theorem say? If a car travels 100 miles in 2 hours, and the position $s(t)$ of the car at time t satisfies the hypotheses of the mean value theorem, can we be sure that there is at least one instant at which the velocity is 50 mph?

7.6.11 Give an example to show that the conclusion of the mean value theorem can fail if we drop the requirement that f be differentiable at every point in (a,b). Give an example to show that the conclusion can fail if we drop the requirement of continuity at the endpoints of the interval.

7.6.12 Suppose that f is differentiable on $[0,\infty)$ and that

$$\lim_{x\to\infty} f'(x) = C.$$

Determine

$$\lim_{x\to\infty} [f(x+a) - f(x)].$$

SEE NOTE 189

7.6.13 Suppose that f is continuous on $[a,b]$ and differentiable on (a,b). If

$$\lim_{x\to a+} f'(x) = C$$

what can you conclude about the right-hand derivative of f at a?

SEE NOTE 190

7.6.14 Suppose that f is continuous and that

$$\lim_{x\to x_0} f'(x)$$

exists. What can you conclude about the differentiability of f? What can you conclude about the continuity of f'?

SEE NOTE 191

7.6.15 Let $f : [0,\infty) \to \mathbb{R}$ so that f' is decreasing and positive. Show that the series

$$\sum_{i=1}^{\infty} f'(i)$$

is convergent if and only if f is bounded.

SEE NOTE 192

7.6.16 Prove a second-order version of the mean value theorem.

> *Let f be continuous on $[a,b]$ and twice differentiable on (a,b). Then there exists $c \in (a,b)$ such that*
>
> $$f(b) = f(a) + (b-a)f'(a) + (b-a)^2 \frac{f''(c)}{2!}.$$

SEE NOTE 193

7.6.17 Determine all functions $f : \mathbb{R} \to \mathbb{R}$ that have the property that

$$f'\left(\frac{x+y}{2}\right) = \frac{f(x) - f(y)}{x-y}$$

for every $x \neq y$.

7.6.18 A function is said to be *smooth* at a point x if

$$\lim_{h\to 0} \frac{f(x+h) + f(x-h) - 2f(x)}{h^2} = 0.$$

Show that a smooth function need not be continuous. Show that if f'' is continuous at x, then f is smooth at x.

SEE NOTE 194

7.6.3 Cauchy's Mean Value Theorem

✂ *Enrichment section. May be omitted.*

We can generalize the mean value theorem to curves given parametrically. Suppose f and g are continuous on $[a,b]$ and differentiable on (a,b). Consider the curve given parametrically by

$$x = g(t), \qquad y = f(t) \qquad (t \in [a,b]).$$

As t varies over the interval $[a,b]$, the point (x,y) traces out a curve C joining the points $(g(a),f(a))$ and $(g(b),f(b))$. If $g(a) \neq g(b)$, the slope of the chord determined by these points is

$$\frac{f(b)-f(a)}{g(b)-g(a)}.$$

Cauchy's form of the mean value theorem asserts that there is a point (x,y) on C at which the tangent is parallel to the chord in question. We state and prove this theorem.

Theorem 7.21 (Cauchy Mean Value Theorem) *Let f and g be continuous on $[a,b]$ and differentiable on (a,b). Then there exists $c \in (a,b)$ such that*

$$[f(b)-f(a)]g'(c) = [g(b)-g(a)]f'(c). \tag{12}$$

Proof. Let

$$\phi(x) = [f(b)-f(a)]g(x) - [g(b)-g(a)]f(x).$$

Then ϕ is continuous on $[a,b]$ and differentiable on (a,b). Furthermore,

$$\phi(a) = f(b)g(a) - f(a)g(b) = \phi(b).$$

By Rolle's theorem, there exists $c \in (a,b)$ for which $\phi'(c) = 0$. It is clear that this point c satisfies (12). ■

Exercises

7.6.19 Use Cauchy's mean value theorem to prove any simple version of L'Hôpital's rule that you can remember from calculus.

7.6.20 Show that the conclusion of Cauchy's mean value can be put into determinant form as

$$\begin{vmatrix} f(a) & g(a) & 1 \\ f(b) & g(b) & 1 \\ f'(c) & g'(c) & 0 \end{vmatrix} = 0.$$

7.6.21 Formulate and prove a generalized version of Cauchy's mean value whose conclusion is the existence of a point c such that

$$\begin{vmatrix} f(a) & g(a) & h(a) \\ f(b) & g(b) & h(b) \\ f'(c) & g'(c) & h'(c) \end{vmatrix} = 0.$$

SEE NOTE 195

7.7 Monotonicity

In elementary calculus one learns that if $f' \geq 0$ on an interval I, then f is nondecreasing on I. We use this and related results for a variety of purposes: sketching graphs of functions, locating extrema, etc. In this section we take a closer look at what's involved. We recall some definitions.

Definition 7.22: Let f be real valued on an interval I.

1. If $f(x_1) \leq f(x_2)$ whenever x_1 and x_2 are points in I with $x_1 < x_2$, we say f is *nondecreasing* on I.

2. If the strict inequality $f(x_1) < f(x_2)$ holds, we say f is *increasing*.

A similar definition was given for *nonincreasing* and *decreasing* functions.

Note. Some authors prefer the terms "increasing" and "strictly increasing" for what we would call nondecreasing and increasing. This has the unfortunate result that constant functions are then considered to be both increasing and decreasing. According to our definition we must say that they are both nondecreasing and nonincreasing, which sounds more plausible—if something stays constant it is neither going up nor going down). The disadvantage of our usage is the discomfort you may at first feel in using the terms (which disappears with practice). It is always safe to say "strictly increasing" for increasing even though it is redundant according to the definition.

By a monotonic function we mean a function that is increasing, decreasing, nondecreasing, or nonincreasing.

The theorems involving monotonicity of functions that one encounters in elementary calculus usually are stated for differentiable functions. But a monotonic function need not be differentiable, or even continuous.

Example 7.23: For example, if

$$f(x) = \begin{cases} x, & \text{for } x < 0 \\ x+1, & \text{for } x \geq 0, \end{cases}$$

then f is increasing on $\mathbb{R}$, but is not continuous at $x = 0$. (For more on discontinuities of monotonic functions, see Section 5.9.2.) ◀

Let us now address the role of the derivative in the study of monotonicity. We prove a familiar theorem that is the basis for many calculus applications. Note that the proof is an easy consequence of the mean value theorem.

Theorem 7.24: *Let f be differentiable on an interval I.*

(i) *If $f'(x) \geq 0$ for all $x \in I$, then f is nondecreasing on I.*

(ii) *If $f'(x) > 0$ for all $x \in I$, then f is increasing on I.*

(iii) *If $f'(x) \leq 0$ for all $x \in I$, then f is nonincreasing on I.*

(iv) *If $f'(x) < 0$ for all $x \in I$, then f is decreasing on I.*

(v) *If $f'(x) = 0$ for all $x \in I$, then f is constant on I.*

Proof. To prove (i), let $x_1, x_2 \in I$ with $x_1 < x_2$. By the mean value theorem (7.20) there exists $c \in (x_1, x_2)$ such that

$$f(x_2) - f(x_1) = f'(c)(x_2 - x_1).$$

If $f'(c) \geq 0$, then $f(x_2) \geq f(x_1)$. Thus, if $f'(x) \geq 0$ for all $x \in I$, f is nondecreasing on I.

Parts (ii), (iii) and (iv) have similar arguments, and (v) follows immediately from parts (i) and (iii). ■

Exercises

7.7.1 Establish the inequality $e^x \leq \frac{1}{1-x}$ for all $x < 1$.

SEE NOTE 196

7.7.2 Suppose that f and g are differentiable functions such that $f' = g$ and $g' = -f$. Show that there exists a number C with the property that

$$[f(x)]^2 + [g(x)]^2 = C$$

for all x.

7.7.3 Suppose f is continuous on (a,c) and $a < b < c$. Suppose also that f is differentiable on (a,b) and on (b,c). Prove that if $f' < 0$ on (a,b) and $f' > 0$ on (b,c), then f has a minimum at b.

SEE NOTE 197

7.7.4 The hypotheses of Theorem 7.24 require that f be differentiable on all of the interval I. You might think that a positive derivative at a single point also implies that the function is increasing, at least in a neighborhood of that point. This is not true. Consider the function

$$f(x) = \begin{cases} x/2 + x^2 \sin x^{-1}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0. \end{cases}$$

(a) Show that the function $g(x) = x^2 \sin x^{-1}$ ($g(0) = 0$) is everywhere differentiable and that $g'(0) = 0$.

(b) Show that g' is discontinuous at $x = 0$ and that g' takes on values close to ± 1 arbitrarily near 0.

(c) Show that f' takes on both positive and negative values in every neighborhood of zero.

(d) Show that $f'(0) = \frac{1}{2} > 0$ but that f is not increasing in any neighborhood of zero.

(e) Prove that if a function F is differentiable on a neighborhood of x_0 with $F'(x_0) > 0$ and F' is *continuous* at x_0, then F is increasing on some neighborhood of x_0.

(f) Why does the example $f(x)$ given here not contradict part (e)?

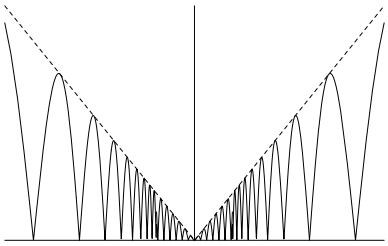


Figure 7.9. Graph of $f(x) = |x\cos x^{-1}|$.

7.7.5 Let f be differentiable on $[0,\infty)$ and suppose that $f(0) = 0$ and that the derivative f' is an increasing function on $[0,\infty)$. Show that

$$\frac{f(x)}{x} < \frac{f(y)}{y}$$

for all $0 < x < y$.

SEE NOTE 198

7.7.6 Suppose that $f, g : \mathbb{R} \to \mathbb{R}$ and both have continuous derivatives and the determinant

$$\phi(x) = \begin{vmatrix} f(x) & g(x) \\ f'(x) & g'(x) \end{vmatrix}$$

is never zero. Show that between any two zeros of f there must be a zero of g.

SEE NOTE 199

7.8 Dini Derivates

✂ *Advanced section. May be omitted.*

We observed in Example 7.4 that the function $f(x) = |x|$ does not have a derivative at the point $x = 0$ but does have the one-sided derivatives $f'_+(0) = 1$ and $f'_-(0) = -1$. It is not difficult to construct continuous functions that don't have even one-sided derivatives at a point.

Example 7.25: Consider the function

$$f(x) = \begin{cases} |x| \left|\cos x^{-1}\right|, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0. \end{cases}$$

(See Figure 7.25). Since $\left|\cos x^{-1}\right| \leq 1$ for all $x \neq 0$,

$$\lim_{x\to 0} f(x) = 0 = f(0)$$

so f is continuous at $x = 0$. It is clear that f is continuous at all other points in $\mathbb{R}$, so f is a continuous function.

The oscillatory behavior of f is such that the sets

$$\left\{x : \left|\cos x^{-1}\right| = 1\right\} \quad \text{and} \quad \left\{x : \left|\cos x^{-1}\right| = 0\right\}$$

both have zero as a two-sided limit point. Thus each of the sets

$$\{x : f(x) = |x|\} \quad \text{and} \quad \{x : f(x) = 0\}$$

has zero as two-sided limit point. Inspection of the difference quotient reveals that

$$\limsup_{x\to 0+} \frac{f(x)-f(0)}{x-0} = 1, \text{ while } \liminf_{x\to 0+} \frac{f(x)-f(0)}{x-0} = 0,$$

so f'_+ does not exist at $x = 0$. Similarly, $f'_-(0)$ does not exist. The limits that are required to exist for f to have a derivative, or a one-sided derivative, don't exist at $x = 0$. ◀

Example 7.26: A function defined on an interval I may fail to have a derivative, even a one-sided derivative, at every point. Let

$$g(x) = \begin{cases} 0, & \text{if } x \text{ is rational,} \\ 1, & \text{if } x \text{ is irrational.} \end{cases}$$

Since g is everywhere discontinuous on both sides, g has no derivative and no one-sided derivative at any point. ◀

There are, also, *continuous* functions that fail to have a one-sided derivative, finite or infinite, at even a single point. Such functions are difficult to construct, the first construction having been given by Besicovitch in 1925.

Now the derivative, when it exists, plays an important role in analysis, and it is useful to have a substitute when it doesn't exist. Many good substitutes have been developed for certain situations. Perhaps the simplest such substitutes are the Dini derivates. These exist at every point for every function defined on an open interval. They are named after the Italian mathematician Ulisse Dini (1845–1918).

Definition 7.27: Let f be real valued in a neighborhood of x_0. We define the four *Dini derivates* of f at x_0 by

1. [Upper right Dini derivate]

$$D^+ f(x_0) = \limsup_{x\to x_0+} \frac{f(x)-f(x_0)}{x-x_0}$$

2. [Lower right Dini derivate]

$$D_+ f(x_0) = \liminf_{x\to x_0+} \frac{f(x)-f(x_0)}{x-x_0}$$

3. [Upper left Dini derivate]

$$D^- f(x_0) = \limsup_{x\to x_0-} \frac{f(x)-f(x_0)}{x-x_0}$$

4. [Lower left Dini derivate]

$$D_-f(x_0) = \liminf_{x\to x_0-} \frac{f(x)-f(x_0)}{x-x_0}.$$

Example 7.28: For the function $f(x) = |x|\left|\cos x^{-1}\right|$, $f(0) = 0$, we calculate that

$$D^+f(0) = 1,\ D_+f(0) = 0,\ D^-f(0) = 0,\ D_-f(0) = -1.$$

Elsewhere $f'(x)$ exists and all four Dini derivatives have that value. ◀

Example 7.29: The function

$$g(x) = \begin{cases} 0, & \text{if } x \text{ is rational,} \\ 1, & \text{if } x \text{ is irrational.} \end{cases}$$

has at every rational x

$$D^+g(x) = 0, D_+g(x) = -\infty, D^-g(x) = \infty, D_-g(x) = 0.$$

For x irrational there are similar values for the Dini derivates (see Exercise 7.8.1a). ◀

It is easy to check that a function f has a derivative at a point x_0 if and only if all four Dini derivates are equal at that point, and a one-sided derivative at x_0 if the two Dini derivates from that side are equal (see Exercise 7.8.2).

We end this section with an illustration of the way in which knowledge about a Dini derivate can substitute for that of the ordinary derivative. We prove a theorem about monotonicity. You are familiar with the fact that if f is differentiable on an interval $[a,b]$ and $f'(x) > 0$ for all $x \in [a,b]$, then f is an increasing function on $[a,b]$. (We provided a formal proof in Section 7.7.)

Here is a generalization of that theorem.

Theorem 7.30: *Let f be continuous on $[a,b]$. If $D^+f(x) > 0$ at each point $x \in [a,b)$, then f is increasing on $[a,b]$.*

Proof. Let us first show that f is nondecreasing on $[a,b]$. We prove this by contradiction. If f fails to be nondecreasing on $[a,b]$, there exist points c and d such that $a \le c < d \le b$ and $f(c) > f(d)$. Let y be any point in the interval $(f(d), f(c))$.

Since f is continuous on $[a,b]$, it possesses the intermediate value property. Thus from Theorem 5.53 [or more precisely from the version of that theorem given as Exercise 5.8.8(a)] there exists a point $t \in (c,d)$ such that $f(t) = y$. Thus the set

$$\{x : f(x) = y)\} \cap [c,d]$$

is nonempty. Let

$$x_0 = \sup\{x : c \le x \le d \text{ and } f(x) = y\}.$$

Now, $f(d) < y$ and f is continuous, from which it follows that $x_0 < d$. Thus $f(x) < y$ for $x \in (x_0, d]$. Furthermore, the set $\{x : f(x) = y\}$ is closed (because f is continuous), so $f(x_0) = y$.

But this implies that $D^+f(x_0) \leq 0$.This contradicts our hypothesis that $D^+f(x) > 0$ for all $x \in [a,b)$. This contradiction completes the proof that f is nondecreasing.

Now we wish to show that it is in fact increasing. If not, then there must be some subinterval in which the function is constant. But at every point interior to that interval we would have $f'(x) = 0$ and so it would be impossible for $D^+f(x) > 0$ at such points. ■

Exercises

7.8.1 Calculate the four Dini derivates for each of the following functions at the given point.

(a)
$$g(x) = \begin{cases} 1, & \text{if } x \text{ is rational} \\ 0, & \text{if } x \text{ is irrational} \end{cases}$$
for $x = \pi$.

(b) $h(x) = x\sin x^{-1}$ $(h(0) = 0)$ at $x = 0$

(c) $f(x) = x\sin x^{-1}$ $(f(0) = 5)$ at $x = 0$

(d)
$$u(x) = \begin{cases} x^2, & \text{if } x \text{ is rational} \\ 0, & \text{if } x \text{ is irrational} \end{cases}$$
at $x = 0$ and at $x = 1$

7.8.2 Prove that f has a derivative at x_0 if and only if
$$D^+f(x_0) = D_+f(x_0) = D^-f(x_0) = D_-f(x_0).$$
In that case, $f'(x_0)$ is the common value of the Dini derivates at x_0. (We assume that f is defined in a neighborhood of x_0.)

7.8.3 **(Derived Numbers)** The Dini derivates are sometimes called "extreme unilateral derived numbers." Let $\lambda \in [-\infty, \infty]$. Then λ is a *derived number* for f at x_0 if there exists a sequence $\{x_k\}$ with $\lim_{k\to\infty} x_k = x_0$ such that
$$\lambda = \lim_{k\to\infty} \frac{f(x_k) - f(x_0)}{x_k - x_0}.$$

(a) For the function $f(x) = \left|x\cos x^{-1}\right|$, $f(0) = 0$, show that every number in the interval $[-1,1]$ is a derived number for f at $x = 0$. Show that the two extreme derived numbers from the right are 0 and 1, and the two from the left are -1 and 0.

(b) Show that a function has a derivative at a point if and only if all derived numbers at that point coincide.

(c) Let $f : \mathbb{R} \to \mathbb{R}$ and let $x_0 \in \mathbb{R}$. Prove that if f is continuous on $\mathbb{R}$, then the set of derived numbers of f at x_0 consists of either one or two closed intervals (that might be degenerate or unbounded). Give examples to illustrate the various possibilities.

7.8.4 Let $f, g : \mathbb{R} \to \mathbb{R}$.

(a) Prove that $D^+(f+g)(x) \leq D^+f(x) + D^+g(x)$.

(b) Give an example to illustrate that the inequality in (a) can be strict.

(c) State and prove the analogue of part (a) for the lower right derivate D_+f.

7.8.5 Generalize Theorem 7.18 to the following:

If f achieves a local maximum at x_0, then $D^+f(x_0) \leq 0$ and $D_-f(x_0) \geq 0$.

Illustrate the result with a function that is not differentiable at x_0.

7.8.6 Prove a variant of Theorem 7.30 that assumes that, for all x in $[a,b)$ except for x in some countable set, the Dini derivate $D^+f(x) > 0$.

7.8.7 Prove a variant of Theorem 7.30: If f is continuous and $D^+f(x) \geq 0$ for all $x \in [a,b)$, then f is nondecreasing on $[a,b]$.
SEE NOTE 200

7.8.8 Prove yet another (more subtle) variant of Theorem 7.30: If f is continuous and $D^+f(x) > 0$ for all $x \in [a,b)$ except for x in some countable set, then f is increasing on $[a,b]$.

7.8.9 Prove that no continuous function can have $D^+f(x) = \infty$ for all $x \in \mathbb{R}$. Give an example of a function $f : \mathbb{R} \to \mathbb{R}$ such that $D^+f(x) = \infty$ for all $x \in \mathbb{R}$.
SEE NOTE 201

7.8.10 Show that the set

$$\{x : D^+f(x) < D_-f(x)\}$$

cannot be uncountable. Give an example of a function f such that $D^+f < D_-f$ on an infinite set.
SEE NOTE 202

7.9 The Darboux Property of the Derivative

Suppose f is differentiable on an interval $[a,b]$. We argued in the proof of Rolle's theorem (7.19) that if $f(a) = f(b)$, then there exists a point $c \in (a,b)$ at which f achieves an extremum. At this point c we have $f'(c) = 0$.

A different hypothesis can lead to the same conclusion. Suppose f is differentiable on $[a,b]$ and $f'(a) < 0 < f'(b)$ (or $f'(b) < 0 < f'(a)$). Once again, the extreme value f achieves must occur at a point c in the *interior* of $[a,b]$, (why?), and at this point we must have $f'(c) = 0$. This observation is a special case of the following theorem first proved by Darboux in 1875.

Theorem 7.31: *Let f be differentiable on an interval I. Suppose $a,b \in I$, $a < b$, and $f'(a) \neq f'(b)$. Let γ be any number between $f'(a)$ and $f'(b)$. Then there exists $c \in (a,b)$ such that $f'(c) = \gamma$.*

Proof. Let $g(x) = f(x) - \gamma x$. If $f'(a) < \gamma < f'(b)$, then $g'(a) = f'(a) - \gamma < 0$ and $g'(b) = f'(b) - \gamma > 0$. The discussion preceding the statement of the theorem shows that there exists $c \in (a,b)$ such that $g'(c) = 0$. For this c we have

$$f'(c) = g'(c) + \gamma = \gamma,$$

completing the proof for the case $f'(a) < f'(b)$.

The proof when $f'(a) > f'(b)$ is similar. ■

You might have noted that Theorem 7.31 is exactly the statement that the derivative of a differentiable function has the Darboux property (i.e., the intermediate value property) that we established for continuous functions in Section 5.8. The derivative f' of a differentiable function f need not be continuous, of course. The result does imply, however, that f' cannot have jump discontinuities and cannot have removable discontinuities.

Both the mean value theorem and Theorem 7.31 give information about the range of the derivative f' of a differentiable function f. The mean value theorem implies that the range of f' includes all slopes of chords determined by the graph of f on the interval of definition of f. Theorem 7.31 tells us that this range is actually an interval. This interval may be unbounded and, if bounded, may or may not contain its endpoints. (See Exercise 7.9.1.)

Derivative of an Inverse Function Theorem 7.31 allows us to establish a familiar theorem about differentiating inverse functions.

Theorem 7.32: *Suppose f is differentiable on an interval I and for each $x \in I$, $f'(x) \neq 0$. Then*

(i) *f is one-to-one on I,*

(ii) *f^{-1} is differentiable on $J = f(I)$,*

(iii) $(f^{-1})'(f(x)) = \dfrac{1}{f'(x)}$ *for all $x \in I$.*

Proof. By Theorem 7.31 either $f'(x) > 0$ for all $x \in I$ or $f'(x) < 0$ for all $x \in I$. In either case, f is either increasing or decreasing on I, and is thus one-to-one, establishing (i).

To verify (ii) and (iii), observe first that f^{-1} is continuous, since f is continuous and strictly monotonic (see Exercise 5.9.16). Let $y_0 \in J$ and let $x_0 = f^{-1}(y_0)$. We wish to show that $(f^{-1})'(y_0)$ exists and has value $1/(f'(x_0))$. For $x \in I$, write $y = f(x)$, so $x = f^{-1}(y)$.

Consider the difference quotient

$$\frac{f^{-1}(y) - f^{-1}(y_0)}{y - y_0} = \frac{x - x_0}{f(x) - f(x_0)}.$$

As $y \to y_0$, $x \to x_0$, because the function f^{-1} is continuous. Thus

$$\lim_{y \to y_0} \frac{f^{-1}(y) - f^{-1}(y_0)}{y - y_0} = \lim_{x \to x_0} \frac{1}{\left(\frac{f(x) - f(x_0)}{x - x_0}\right)} = \frac{1}{f'(x_0)}.$$

■

Exercises

7.9.1 Let f be differentiable on $[a,b]$ and let $\mathcal{R}(f')$ denote the range of f' on $[a,b]$. Give examples to illustrate that $\mathcal{R}(f')$ can be

(a) a closed interval

(b) an open interval

(c) a half-open interval

(d) an unbounded interval

SEE NOTE 203

7.9.2 Give an example of a differentiable function f such that

$$f'(x_0) \neq \lim_{x \to x_0} f'(x).$$

Show that if f is defined and continuous in a neighborhood of x_0 and if the limit

$$\lim_{x \to x_0} f'(x)$$

exists and is finite, then f is differentiable at x_0 and f' is continuous at x_0.

7.9.3 Most classes of functions we have encountered are closed under the operations of addition and multiplication (e.g., polynomials, continuous functions, differentiable functions). The class of derivatives is closed under addition, but behaves badly with respect to multiplication. Consider, for example, the pair of functions F and G defined on $\mathbb{R}$ by

$$F(x) = x^2 \sin \frac{1}{x^3}, \ (F(0) = 0), \text{ and}$$
$$G(x) = x^2 \cos \frac{1}{x^3}, \ (G(0) = 0).$$

Verify each of the following statements:

(a) F and G are differentiable on $\mathbb{R}$.

(b) The functions FG' and GF' are bounded functions.

(c) $F(x)G'(x) - F'(x)G(x) = \begin{cases} 3, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0. \end{cases}$

(d) At least one of the functions FG' or GF' must fail to be a derivative.

Thus, even the product of a differentiable function F with a derivative G' need not be a derivative.

SEE NOTE 204

7.9.4 Show, in contrast to Exercise 7.9.3, that if a function f has a continuous derivative on $\mathbb{R}$ and g is differentiable, then fg' is a derivative.

SEE NOTE 205

7.9.5 Let f be a differentiable function on an interval $[a,b]$. Show that f' is continuous if and only if the set

$$E_\alpha = \{x : f'(x) = \alpha\}$$

is closed for each real number α.

SEE NOTE 206

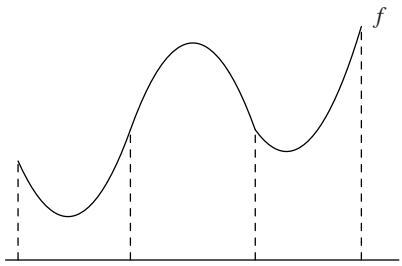


Figure 7.10. Concave up/down/up.

7.9.6 Let $f:[0,1]\to\mathbb{R}$ be a continuous function that is differentiable on $(0,1)$ and with $f(0)=0$ and $f(1)=1$. Show there must exist distinct numbers ξ_1 and ξ_2 in that interval such that

$$f'(\xi_1)f'(\xi_2)=1.$$

7.9.7 Prove or disprove that if $f:\mathbb{R}\to\mathbb{R}$ is differentiable and monotonic, then f' must be continuous on $\mathbb{R}$.

7.10 Convexity

In elementary calculus one studies functions that are concave-up or concave-down on an interval. A knowledge of the intervals on which a function is concave-up or concave-down is useful for such purposes as sketching the graph of the equation $y=f(x)$ and studying extrema of the function (Fig. 7.10).

In the setting of elementary calculus the functions usually have second derivatives on the intervals involved. In that setting we define a function as being concave-up on an interval I if $f''\geq 0$ on I, and concave-down if $f''\leq 0$ on I. Definitions involving the first derivative, but not the second, can also be given: f is concave-up on I if f' is increasing on I, concave-down if f' is decreasing on I. Equivalently, f is concave-up if the graph of f lies "above" (more precisely "not below") each of its tangent lines, concave-down if the graph lies below (not above) each of its tangent lines.

The geometric properties we wish to capture when we say a function is concave-up or concave-down do not depend on differentiability properties. The condition is that the graph should lie below (or above) all its chords. The following definitions make this concept precise. We shall follow the common practice of using the terms "convex" and "concave" in place of the terms "concave-up" and "concave-down."

Definition 7.33: Let f be defined on an interval I. If for all $x_1,x_2\in I$ and $\alpha\in[0,1]$ the inequality

$$f(\alpha x_1+(1-\alpha)x_2)\leq \alpha f(x_1)+(1-\alpha)f(x_2) \tag{13}$$

is satisfied, we say that f is *convex* on I. If the reverse inequality in (13) applies, we say that f is *concave* on I. If the inequalities are strict for all $\alpha\in(0,1)$ we say f is *strictly convex* or *strictly concave* on I.

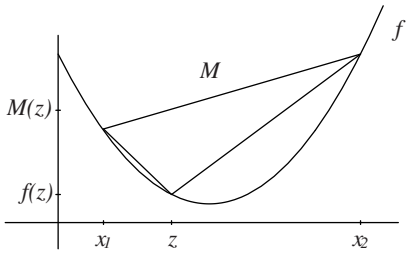


Figure 7.11. Comparison of the three slopes in the inequalities (14).

For example, the function $f(x) = |x|$ is convex, but not strictly convex on $\mathbb{R}$. Strict convexity implies that the graph of f has no line segments in it. Note that the function $f(x) = |x|$ is not differentiable at $x = 0$.

The geometric condition defining convexity does imply a great deal of regularity of a function. Our first objective is to address this issue. We begin with some simple geometric considerations.

Suppose f is convex on an open interval I. Let x_1 and x_2 be points in I with $x_1 < x_2$. The chord determined by the points $(x_1, f(x_1))$ and $(x_2, f(x_2))$ defines a linear function M on $[x_1, x_2]$: If $x = \alpha x_1 + (1-\alpha)x_2$, then

$$M(x) = \alpha f(x_1) + (1-\alpha) f(x_2).$$

The definition of "convex" states that

$$f(x) \le M(x)$$

for all $x \in [x_1, x_2]$ and that

$$M(x_1) = f(x_1) \text{ and } M(x_2) = f(x_2).$$

Now let $z \in (x_1, x_2)$ Then

$$\frac{f(z) - f(x_1)}{z - x_1} \le \frac{M(z) - M(x_1)}{z - x_1} = \frac{M(x_2) - M(z)}{x_2 - z} \le \frac{f(x_2) - f(z)}{x_2 - z} \tag{14}$$

(Fig. 7.11).

Thus, the chord determined by f and the points x_1 and x_2 has a slope between the slopes of the chord determined by x_1 and z and the chord determined by z and x_2.

The inequalities (14) have a number of useful consequences:

1. For fixed $x \in I$,
$$(f(x+h) - f(x))/h$$
is a nondecreasing function of h on some interval $(0, \delta)$. Thus
$$\lim_{h \to 0+} \frac{f(x+h) - f(x)}{h} = \inf_{h > 0} \frac{f(x+h) - f(x)}{h}$$

exists or possibly is $-\infty$. That it is in fact finite can be shown by using (14) again to get a finite lower bound, since

$$(f(x')-f(x))/(x'-x) \le (f(x+h)-f(x))/h$$

for any $x' \in I$ with $x' < x$. Thus f has a right-hand derivative $f'_+(x)$ at x. Similarly, f has a finite left-hand derivative at x.

2. If $x, y \in I$ and $x < y$, then
$$f'_+(x) \le f'_+(y).$$
From observation 1 we infer that
$$(f(x+h)-f(x))/h \le (f(y+h)-f(y))/h$$
whenever $h > 0$ and $x+h, y+h$ are in I. Thus f'_+ is a nondecreasing function. Similarly, f'_- is a nondecreasing function.

3. It is also clear from (14) that
$$f'_-(x) \le f'_+(x)$$
for all $x \in I$.

4. f is continuous on I. To see this, observe that since both one-sided derivatives exist at every point the function must be continuous on both sides, hence continuous.

We summarize the preceding discussion as a theorem.

Theorem 7.34: *Let f be convex on an open interval I. Then*

(i) *f has finite left and right derivatives at each point of I. Each of these one-sided derivatives is a nondecreasing function of x on I, and*
$$f'_-(x) \le f'_+(x) \text{ for all } x \in I. \tag{15}$$

(ii) *f is continuous on I.*

Note. If f is convex on a *closed* interval $[a,b]$, some of the results do not apply at the endpoints a and b. (See Exercise 7.10.8.) Note, too, that the corresponding results are valid for concave functions on I, the one-sided derivatives now being nonincreasing functions of x and the inequality in (15) being reversed.

We can now obtain the characterizations of convex functions familiar from elementary calculus.

Corollary 7.35: *Let f be defined on an open interval I.*

(i) *If f is differentiable on I, then f is convex on I if and only if f' is nondecreasing on I.*

(ii) *If f is twice differentiable on I, then f is convex on I if and only if $f'' \ge 0$ on I.*

We leave the verification of Corollary 7.35 as Exercise 7.10.9.

Exercises

7.10.1 Show that a function f is convex on an interval I if and only if the determinant

$$\begin{vmatrix} 1 & x_1 & f(x_1) \\ 1 & x_2 & f(x_2) \\ 1 & x_3 & f(x_3) \end{vmatrix}$$

is nonnegative for any choices of $x_1 < x_2 < x_3$ in the interval I.

7.10.2 If f and g are convex on an interval I, show that any linear combination $\alpha f + \beta g$ is also convex provided α and β are nonnegative.

7.10.3 If f and g are convex functions, can you conclude that the composition $g \circ f$ is also convex?

SEE NOTE 207

7.10.4 Let f be convex on an open interval (a,b). Show that then there are only two possibilities. Either (i) f is nonincreasing or nondecreasing on the entire interval (a,b) or else (ii) there is a number c so that f is nonincreasing on $(a,c]$ and nondecreasing on $[c,b)$.

7.10.5 Suppose f is convex on an open interval I. Prove that f is differentiable except on a countable set.

SEE NOTE 208

7.10.6 Suppose f is convex on an open interval I. Prove that if f is differentiable on I, then f' is continuous on I.

7.10.7 Let f be convex on an open interval that contains the closed interval $[a,b]$. Let

$$M = \max\{f'_+(a), f'_-(b)\}.$$

Show that

$$|f(x) - f(y)| \leq M|x - y|$$

for all x, $y \in [a,b]$.

7.10.8 Theorem 7.34 pertains to functions that are convex on an open interval. Discuss the extent to which the results of the theorem hold when f is convex on a *closed* interval $[a,b]$. In particular, determine whether continuity of f at the endpoints of the interval follows from the definition. Must $f'_+(a)$ and $f'_-(b)$ be finite?

7.10.9 Prove Corollary 7.35.

7.10.10 Let f be convex on an open interval (a,b). Must f be bounded above? Must f be bounded below?

SEE NOTE 209

7.10.11 Let f be convex on an open interval (a,b). Show that f does not have a strict maximum value.

7.10.12 Let f be defined and continuous on an open interval (a,b). Show that f is convex there if and only if there do not exist real numbers α and β such that the function $f(x) + \alpha x + \beta$ has a strict maximum value in (a,b).

7.10.13 ✂ Let $A = \{a_1, a_2, a_3, \dots\}$ be any countable set of real numbers. Let

$$f(x) = \sum_1^\infty \frac{|x - a_k|}{10^k}.$$

Prove that f is convex on $\mathbb{R}$, differentiable on the set $\mathbb{R} \setminus A$, and nondifferentiable on the set A.

SEE NOTE 210

7.10.14 ✂ **(Inflection Points)** In elementary calculus one studies inflection points. The definitions one finds try to capture the idea that at such a point the sense of concavity changes from strict "up to down" or vice versa. Here are three common definitions that apply to differentiable functions. In each case f is defined on an open interval (a,b) containing the point x_0. The point x_0 is an *inflection point* for f if there exists an open interval $I \subset (a,b)$ such that on I

(Definition A) f' increases on one side of x_0 and decreases on the other side.

(Definition B) f' attains a strict maximum or minimum at x_0.

(Definition C) The tangent line to the graph of f at $(x_0, f(x_0))$ lies below the graph of f on one side of x_0 and above on the other side.

(a) Prove that if f satisfies Definition A at x_0, then it satisfies Definition B at x_0.

(b) Prove that if f satisfies Definition B at x_0, then it satisfies Definition C at x_0.

(c) Give an example of a function satisfying Definition B at x_0, but not satisfying Definition A.

(d) Give an example of an infinitely differentiable function satisfying Definition C at x_0, but not satisfying Definition B.

(e) Which of the three definitions states that the sense of concavity of f is "up" on one side of x_0 and "down" on the other?

SEE NOTE 211

7.10.15 **(Jensen's Inequality)** Let f be a convex function on an interval I, let $x_1, x_2, \dots, x_n$ be points of I and let $\alpha_1, \alpha_2, \dots, \alpha_n$ be positive numbers satisfying

$$\sum_{k=1}^n \alpha_k = 1.$$

Show that

$$f\left(\sum_{k=1}^n \alpha_k x_k\right) \leq \sum_{k=1}^n \alpha_k f(x_k).$$

SEE NOTE 212

7.10.16 Show that the inequality is strict in Jensen's inequality (Exercise 7.10.15) except in the case that f is linear on some interval that contains the points $x_1, x_2, \dots, x_n$.

7.11 L'Hpital's Rule

✂ *Enrichment section. May be omitted.*

Suppose that f and g are defined in a deleted neighborhood of x_0 and that

$$\lim_{x\to x_0} f(x) = A \text{ and } \lim_{x\to x_0} g(x) = B.$$

According to our usual theory of limits, we then have

$$\lim_{x\to x_0} \frac{f(x)}{g(x)} = \frac{\lim_{x\to x_0} f(x)}{\lim_{x\to x_0} g(x)} = \frac{A}{B},$$

unless $B = 0$.

But what happens if $B = 0$, which is often the case? A number of possibilities exist: If $B = 0$ and $A \neq 0$, then the limit does not exist. The most interesting case remains: If both A and B are zero, then the limiting behavior depends on the rates at which $f(x)$ and $g(x)$ approach zero.

Example 7.36: Consider

$$\lim_{x\to 0} \frac{6x}{3x} = \lim_{x\to 0} \frac{6}{3} = 2.$$

Look at this simple example geometrically. For $x \neq 0$, the height $6x$ is twice that of the height $3x$. The straight line $y = 6x$ approaches zero at twice the rate that the line $y = 3x$ does. ◀

Example 7.37: Now consider the slightly more complicated limit

$$\lim_{x\to 0} \frac{f(x)}{g(x)} = \lim_{x\to 0} \frac{6x+x^2}{3x+5x^3}.$$

If we divide the numerator and denominator by $x \neq 0$, we see that the limit is the same as

$$\lim_{x\to 0} \frac{6+x}{3+5x^2}.$$

This last limit can be calculated by our usual elementary methods as equaling $6/3 = 2$. Here, for $x \neq 0$ near zero, the height $f(x) = 6x + x^2$ is approximately $6x$, while the height of $g(x) = 3x + 5x^3$ is approximately $3x$, that is, the desired ratio is approximately 2. Again, the numerator approaches zero at about twice the rate that the denominator does.

We can be more precise by calculating these rates exactly. Let

$$f(x) = 6x + x^2 \text{ and } g(x) = 3x + 5x^3.$$

Then

$$\begin{aligned} f'(x) &= 6 + 2x, \quad f'(0) = 6 \\ g'(x) &= 3 + 5x^2, \quad g'(0) = 3. \end{aligned}$$

This makes precise our statement that the numerator approaches zero twice as fast as the denominator does. (See Figure 7.12 where there is an illustration showing the graphs of the functions f and g compared to the lines $y = 6x$ and $y = 3x$.) ◀

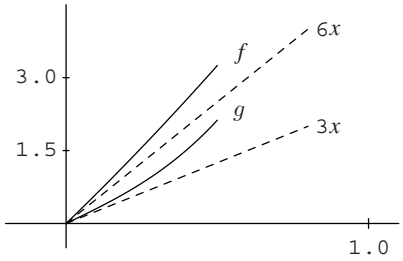


Figure 7.12. Comparison of the rates in Example 7.37.

Let us try to generalize from these two examples. Suppose f and g are differentiable in a neighborhood of $x = a$ and that $f(a) = g(a) = 0$. Consider the following calculations and what conditions on f and g are required to make them valid.

$$\frac{f(x)}{g(x)} = \frac{f(x)-f(a)}{g(x)-g(a)} = \frac{\left(\frac{f(x)-f(a)}{x-a}\right)}{\left(\frac{g(x)-g(a)}{x-a}\right)} \xrightarrow{x\to a} \frac{f'(a)}{g'(a)} = \lim_{x\to a}\frac{f'(x)}{g'(x)}. \tag{16}$$

If these calculations are valid, they show that under these assumptions ($f(a) = g(a) = 0$ and both $f'(a)$ and $g'(a)$ exist) we should be able to claim that

$$\lim_{x\to a}\frac{f(x)}{g(x)} = \lim_{x\to a}\frac{f'(x)}{g'(x)}.$$

You should check the various conditions that must be met to justify the calculations: $g(x)$ cannot equal zero at any point of the neighborhood in question (other than a); nor can $g(x) = g(a)$, (for $x \neq a$); $f(a)$ and $g(a)$ must equal zero (for the first equality), and f'/g' must be continuous at $x = a$ (for the last equality).

The calculations (16) provide a simple proof of a rudimentary form for a method of computing limits known as L'Hôpital's rule. We say "rudimentary" because some of the conditions we assumed are not needed for the conclusion

$$\lim_{x\to a}\frac{f(x)}{g(x)} = \lim_{x\to a}\frac{f'(x)}{g'(x)}.$$

7.11.1 L'Hpital's Rule: $\frac{0}{0}$ Form

✂ *Enrichment section. May be omitted.*

Our first theorem provides a version of the rule identical with our introductory remarks but under weaker assumptions.

Theorem 7.38 (L'Hôpital's Rule: $\frac{0}{0}$ Form) *Suppose that the functions f and g are differentiable in a deleted neighborhood N of $x = a$. Suppose*

(i) $\lim_{x\to a} f(x) = 0$,

(ii) $\lim_{x\to a} g(x) = 0$,

(iii) *For every* $x \in N$, $g'(x) \neq 0$, *and*

(iv) $\lim_{x\to a} \frac{f'(x)}{g'(x)}$ *exists.*

Then $\lim_{x\to a} \dfrac{f(x)}{g(x)} = \lim_{x\to a} \dfrac{f'(x)}{g'(x)}$.

Proof. Our hypotheses do not require f and g to be defined at $x = a$. But we can in any case define (or redefine) f and g at $x = a$ by $f(a) = g(a) = 0$. Because of assumptions (i) and (ii), this results in continuous functions defined on the full neighborhood $N \cup \{a\}$ of the point $x = a$. We can now apply Cauchy's form of the mean value theorem (7.21).

Suppose $x \in N$ and $a < x$. By Theorem 7.21 there exists $c = c_x$ in (a,x) such that

$$[f(x) - f(a)]g'(c_x) = [g(x) - g(a)]f'(c_x). \tag{17}$$

Since $f(a) = g(a) = 0$, (17) becomes

$$f(x)g'(c_x) = g(x)f'(c_x). \tag{18}$$

Equation (18) is valid for $x > a$ in N. We would like to express (18) in the form

$$\frac{f(x)}{g(x)} = \frac{f'(c_x)}{g'(c_x)}. \tag{19}$$

To justify (19) we show that $g(x)$ is never zero in $N \cap \{x : x > a\}$. (That $g'(c_x)$ is never zero in N is our hypothesis (iii).) If for some $x \in N$, $x > a$, we have $g(x) = 0$, then by Rolle's theorem there would exist a point $t \in (a,x)$ such that $g'(t) = 0$, contradicting hypothesis (ii). Thus equation (19) is valid for all $N \cap \{x : x > a\}$. A similar argument shows that if $x \in N$, $x < a$, then there exists $c_x \in (x,a)$ such that (19) holds.

Now as $x \to a$, c_x also approaches a, since c_x is between a and x. Thus

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(c_x)}{g'(c_x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)},$$

since the last limit exists by hypothesis (iv). ■

Note. Observe that we did not require f to be defined at $x = a$, nor did we require that f'/g' be continuous at $x = a$. It is also important to observe that L'Hôpital's rule does *not* imply that, under hypotheses (i), (ii), and (iii) of Theorem 7.38, if $\lim_{x\to a} f(x)/g(x)$ exists, then $\lim_{x\to a} f'(x)/g'(x)$ must also exist. Exercise 7.11.5 provides an example to illustrate this.

Example 7.39: Let us use L'Hôpital's rule to evaluate

$$\lim_{x\to 0} \ln(1+x)/x.$$

Let $f(x) = \ln(1+x)$, $g(x) = x$. Then

$$\lim_{x\to 0} f(x) = \lim_{x\to 0} g(x) = 0,\ f'(x) = \frac{1}{1+x},\ \text{and}\ g'(x) = 1.$$

Thus

$$\lim_{x\to 0}\frac{\ln(1+x)}{x}=\lim_{x\to 0}\frac{1}{1+x}=1.$$

◀

We refer to this theorem as the "$\frac{0}{0}$ form" for obvious reasons. There is also a version of the form $\frac{\infty}{\infty}$ (see Theorem 7.42). In addition, other modifications are possible. The point a can be replaced with $a=\infty$ or $a=-\infty$, (Theorem 7.41), and the results are valid for one-sided limits. (Our proof of Theorem 7.38 actually established that fact since we considered the case $x>a$ and $x<a$ separately.) Various other "indeterminate forms," ones for which the limit depends on the rates at which component parts approach their separate limits, can be manipulated to make use of L'Hôpital's rule possible.

Here is an example in which the forms "1^{∞}" and "$1^{-\infty}$" come into play. Observe that the function whose limit we wish to calculate is of the form $f(x)^{g(x)}$ where $f(x)\to 1$ as $x\to a$ but $g(x)\to\infty$ as $x\to a+$ and $g(x)\to-\infty$ as $x\to a-$.

Example 7.40: Evaluate $\lim_{x\to 0}(1+x)^{2/x}$. This expression is of the form 1^{∞} (when $x>0$). To calculate $\lim_{x\to 0}(1+x)^{2/x}$, write

$$y=(1+x)^{2/x}, z=\ln y=\frac{2}{x}\ln(1+x).$$

Now the numerator and denominator of the function z satisfy the hypotheses of L'Hôpital's rule. Thus

$$\lim_{x\to 0}z=\lim_{x\to 0}\frac{2\ln(1+x)}{x}=\lim_{x\to 0}\frac{2}{1+x}=2.$$

Since $\lim_{x\to 0}z=2$, $\lim_{x\to 0}y=e^2$. ◀

7.11.2 L'Hpital's Rule as $x\to\infty$

✂ *Enrichment section. May be omitted.*

We proved Theorem 7.38 under the assumption that $a\in\mathbb{R}$, but the theorem is valid when $a=-\infty$ or $a=+\infty$. In this case we are, of course, dealing with one-sided limits. As before, the relation

$$\lim_{x\to\infty}\frac{f'(x)}{g'(x)}=L$$

implies something about relative rates of growth of the functions $f(x)$ and $g(x)$ as $x\to\infty$. We can base a proof of the versions of L'Hôpital's rule that have $a=\infty$ (or $-\infty$) on Theorem 7.38 by a simple transformation.

Theorem 7.41: *Let f,g be differentiable on some interval $(-\infty,b)$. Suppose*

(i) $\lim_{x\to-\infty}f(x)=0$,

(ii) $\lim_{x\to-\infty}g(x)=0$,

(iii) *For every $x\in(-\infty,b)$, $g'(x)\neq 0$, and*

(iv) $\lim_{x\to-\infty}\frac{f'(x)}{g'(x)}$ *exists.*

Then

$$\lim_{x\to-\infty}\frac{f(x)}{g(x)}=\lim_{x\to-\infty}\frac{f'(x)}{g'(x)}.$$

A similar result holds when we replace ∞ by $-\infty$ in the hypotheses.

Proof. Let $x=-1/t$. Then, as $t\to 0+$, $x\to-\infty$ and vice-versa. Define functions F and G by

$$F(t)=f\left(-\frac{1}{t}\right)\ \text{ and }\ G(t)=g\left(-\frac{1}{t}\right).$$

Both functions F and G are defined on some interval $(0,\delta)$. We verify easily that

$$\lim_{t\to0+}F(t)=\lim_{t\to0+}G(t)=0$$

and that

$$\lim_{t\to0+}\frac{F'(t)}{G'(t)}=\lim_{x\to-\infty}\frac{f'(x)}{g'(x)}. \tag{20}$$

Using Theorem 7.38, we infer

$$\lim_{t\to0+}\frac{F'(t)}{G'(t)}=\lim_{t\to0+}\frac{F(t)}{G(t)}=\lim_{t\to0+}\frac{f(-\frac{1}{t})}{g(-\frac{1}{t})}=\lim_{x\to-\infty}\frac{f(x)}{g(x)}. \tag{21}$$

The result follows from (20) and (21) ■

7.11.3 L'Hpital's Rule: $\frac{\infty}{\infty}$ Form

✂ *Enrichment section. May be omitted.*

When $f(x)\to\infty$ and $g(x)\to\infty$ as $x\to a$ we obtain the indeterminate form $\frac{\infty}{\infty}$. L'Hôpital's theorem then takes the form given in Theorem 7.42. Note, however, that we don't require $f(x)\to\infty$ in our hypotheses, or even that $f(x)$ approaches any limit.

Theorem 7.42: *Let f and g be differentiable on a deleted neighborhood N of $x=a$. Suppose that*

(i) $\lim_{x\to a}g(x)=\infty$.

(ii) *For every $x\in N$ $g'(x)\neq0$.*

(iii) $\lim_{x\to a}f'(x)/g'(x)$ *exists.*

Then

$$\lim_{x\to a}\frac{f(x)}{g(x)}=\lim_{x\to a}\frac{f'(x)}{g'(x)}.$$

The analogous statements are valid if $a=\pm\infty$ or if $\lim_{x\to a}g(x)=-\infty$.

Proof. We prove the main part of Theorem 7.42 under the assumption that

$$\lim_{x\to a}f'(x)/g'(x)$$

is finite. The case that the limit is infinite as well as variants are left as Exercises 7.11.6 and 7.11.7. It suffices to consider the case of right-hand limits, the proof for left-hand limits being similar. Let

$$L = \lim_{x \to a+} f'(x)/g'(x).$$

We will show that if $p < L < q$, then there exists $\delta > 0$ such that

$$p < f(x)/g(x) < q$$

for $x \in (a, a+\delta)$. Since p and q are arbitrary (subject to the restriction $p < L < q$), we can then conclude

$$\lim_{x \to a+} f(x)/g(x) = L$$

as required.

Choose $r \in (L, q)$. By (iii) and the definition of L there exists δ_1 such that $f'(x)/g'(x) < r$ whenever $x \in (a, a+\delta_1)$. If $a < x < y < a+\delta_1$, then we infer from Theorem 7.21, Cauchy's form of the mean value theorem, and our assumption (ii) that there exists $c \in (x, y)$ such that

$$\frac{f(x) - f(y)}{g(x) - g(y)} = \frac{f'(c)}{g'(c)} < r. \tag{22}$$

Fix y in (22). Since $\lim_{x \to a+} g(x) = \infty$, there exists $\delta_2 > 0$ such that $a + \delta_2 < y$ and such that $g(x) > g(y)$ and $g(x) > 0$ if $a < x < a + \delta_2$. We then have

$$(g(x) - g(y))/g(x) > 0$$

for $x \in (a, a+\delta_2)$, so we can multiply both sides of the inequality (22) by $(g(x) - g(y))/g(x)$, obtaining

$$\frac{f(x)}{g(x)} < r - r\frac{g(y)}{g(x)} + \frac{f(y)}{g(x)} \text{ for } x \in (a, a+\delta_2). \tag{23}$$

Now let $x \to a+$. Then $g(x) \to \infty$ as $x \to a+$ by assumption (i). Since r, $g(y)$, and $f(y)$ are constants, the second and third terms on the right side of (23) approach zero. It now follows from the inequality $r < q$ that there exists $\delta_3 \in (0, \delta_2)$ such that

$$\frac{f(x)}{g(x)} < q \text{ whenever } a < x < a + \delta_3. \tag{24}$$

In a similar fashion we find a $\delta_4 > 0$ such that

$$\frac{f(x)}{g(x)} > p \text{ whenever } a < x < a + \delta_4.$$

If we let $\delta = \min(\delta_3, \delta_4)$, we have shown that

$$p < \frac{f(x)}{g(x)} < q \text{ whenever } x \in (a, a+\delta).$$

Since p and q were arbitrary numbers satisfying $p < L < q$, our conclusion

$$\lim_{x \to a+} \frac{f(x)}{g(x)} = L = \lim_{x \to a+} \frac{f'(x)}{g'(x)}$$

follows. ■

Exercises

7.11.1 Consider the function $f(x) = (3^x - 2^x)/x$ defined everywhere except at $x = 0$.

(a) What value should be assigned to $f(0)$ in order that f be everywhere continuous?

(b) Does $f'(0)$ exist if this value is assigned to $f(0)$?

(c) Would it be correct to calculate $f'(0)$ by computing instead $f'(x)$ by the usual rules of the calculus and finding $\lim_{x\to 0} f'(x)$.

SEE NOTE 213

7.11.2 Suppose that f and g are defined in a deleted neighborhood of x_0 and that

$$\lim_{x\to x_0} f(x) = A \neq 0 \text{ and } \lim_{x\to x_0} g(x) = 0.$$

Show that

$$\lim_{x\to x_0} \left| \frac{f(x)}{g(x)} \right| = \infty.$$

SEE NOTE 214

7.11.3 Discuss the limiting behavior as $x \to 0$ for each of the following functions.

(a) $\dfrac{1}{x}$ (b) $\dfrac{1}{x^2}$

(c) $\dfrac{1}{\sin x}$ (d) $\dfrac{1}{x \sin x^{-1}}$

7.11.4 Evaluate each of the following limits.

(a) $\lim\limits_{x\to 0} \dfrac{e^x - \cos x}{x}$

(b) $\lim\limits_{t\to 0} \dfrac{\sin t - t}{t^3}$

(c) $\lim\limits_{u\to 1} \dfrac{u^5 + 5u - 6}{2u^5 + 8u - 10}$

7.11.5 Let $f(x) = x^2 \sin x^{-1}$, $g(x) = x$. Show that

$$\lim_{x\to 0} \frac{f(x)}{g(x)} = 0$$

but that

$$\lim_{x\to 0} \frac{f'(x)}{g'(x)}$$

does not exist.

7.11.6 The proof we provided for Theorem 7.42 required that $\lim_{x\to a} f'(x)/g'(x)$ be finite. Prove that the result holds if this limit is infinite.

7.11.7 Prove the part of Theorem 7.42 dealing with $a = \pm\infty$ or $\lim_{x\to a} g(x) = -\infty$.

7.11.8 Evaluate the following limits.

(a) $\lim\limits_{x\to\infty} \dfrac{x^3}{e^x}$

(b) $\lim_{x\to\infty} \dfrac{\ln x}{x}$

(c) $\lim_{x\to 0+} x\ln x$

(d) $\lim_{x\to 0+} x^x$

7.11.9 This exercise gives information about the relative rates of increase of certain types of functions. Prove that for each positive number p,

$$\lim_{x\to\infty} \frac{\ln x}{x^p} = \lim_{x\to\infty} \frac{x^p}{e^x} = 0.$$

7.11.10 Give an example of functions f and g defined on $\mathbb{R}$ such that

$$\lim_{x\to\infty} g(x) = \infty,\ \limsup_{x\to\infty} f(x) = \infty,\ \liminf_{x\to\infty} f(x) = -\infty$$

and Theorem 7.42 applies.

SEE NOTE 215

7.12 Taylor Polynomials

✂ *Enrichment section. May be omitted.*

Suppose f is continuous on an open interval I and $c \in I$. The constant function $g(x) = f(c)$ approximates f closely when x is sufficiently close to the point c, but may or may not provide a good approximation elsewhere. If f is differentiable on I, then we see from the mean value theorem (Theorem 7.20) that for each $x \in I$ ($x \neq c$) there exists z between x and c such that

$$f(x) = f(c) + f'(z)(x-c).$$

The expression $R_0(x) = f'(z)(x-c) = f(x) - f(c)$ provides the size of the error obtained in approximating the function f by a constant function $P_0(x) = f(c)$. We can think of this as approximation by a zero-degree polynomial.

We do not expect a constant function to be a good approximation to a given continuous function in general. But our acquaintance with Taylor series (as presented in elementary calculus courses) suggests that if a function is sufficiently differentiable, it can be approximated well by polynomials of sufficiently high degree.

Suppose we wish to approximate f by a polynomial P_n of degree n. In order for the polynomial P_n to have a chance to approximate f well in a neighborhood of a point c, we should require

$$P_n(c) = f(c), P_n'(c) = f'(c), \ldots, P_n^{(n)}(c) = f^{(n)}(c).$$

In that case we at least guarantee that P_n "starts out" with the correct value, the correct rate of change, etc. to give it a chance to approximate f well in some neighborhood I of c. The test however is this. Write

$$f(x) = P_n(x) + R_n(x).$$

Is it true that the "error" or "remainder" $R_n(x)$ is small when $x \in I$?

In order to answer this sort of question, it would be useful to have workable forms for this error term $R_n(x)$. We present two forms for the remainder. The first is due to

Joseph-Louis Lagrange (1736–1813), who obtained Theorem 7.43 in 1797. He used integration methods to prove the theorem. We provide a popular and more modern proof based on the mean value theorem.

Theorem 7.43 (Lagrange) *Let f possess at least $n+1$ derivatives on an open interval I and let $c \in I$. Let*

$$P_n(x) = f(c) + f'(c)(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \cdots + \frac{f^{(n)}(c)}{n!}(x-c)^n$$

and let $R_n(x) = f(x) - P_n(x)$. Then for each $x \in I$ there exists z between x and c ($z = c$ if $x = c$) such that

$$R_n(x) = \frac{f^{(n+1)}(z)}{(n+1)!}(x-c)^{n+1}.$$

Proof. Fix $x \in I$. Then there is a number M (depending on x, of course) such that

$$f(x) = P_n(x) + M(x-c)^{n+1}.$$

We wish to show that $M = (f^{(n+1)}(z))/(n+1)!$ for some z between x and c.

Consider the function g defined on I by

$$\begin{aligned} g(t) &= f(t) - P_n(t) - M(t-c)^{n+1} \\ &= R_n(t) - M(t-c)^{n+1}. \end{aligned}$$

Now P_n is a polynomial of degree at most n, so $P_n^{(n+1)}(t) = 0$ for all $t \in I$. Thus

$$g^{(n+1)}(t) = f^{(n+1)}(t) - (n+1)!M \text{ for all } t \in I. \tag{25}$$

Also, since $f^{(k)}(c) = P_n^{(k)}(c)$ for $k = 1, 2, \ldots, n$, we readily see that

$$g^{(k)}(c) = 0 \text{ for } k = 0, 1, 2, \ldots, n. \tag{26}$$

Suppose now that $x > c$, the case $x < c$ having a similar proof, and the case $x = c$ being obvious. We have chosen M in such a way that $g(x) = 0$ and, by (26), we see that $g(c) = 0$. Thus g satisfies the hypotheses of Rolle's theorem on the interval $[c, x]$. Therefore there exists a point $z_1 \in (c, x)$ such that $g'(z_1) = 0$.

Now apply Rolle's theorem to g' on the interval $[c, z_1]$, obtaining a point $z_2 \in (c, z_1)$ such that $g''(z_2) = 0$.

Continuing in this way we use (26) and Rolle's theorem repeatedly to obtain a point $z_n \in (c, z_{n-1})$ such that $g^{(n)}(z_n) = 0$. Finally, we apply Rolle's theorem to the function $g^{(n)}$ on the interval $[c, z_n]$. We obtain a point $z \in (c, z_n)$ such that $g^{(n+1)}(z) = 0$. From (25) we deduce

$$f^{(n+1)}(z) = (n+1)!M,$$

completing the proof. ■

Note. The function P_n is called the *nth Taylor polynomial for f*. You will recognize P_n as the nth partial sum of the Taylor series studied in elementary calculus. (See also

Chapter 10.) The function R_n is called the *remainder* or *error* function between f and P_n. If P_n is to be a good approximation to f, then R_n must be small in absolute value.

Observe that $P_n(c) = f(c)$ and that

$$P_n^{(k)}(c) = f^{(k)}(c) \text{ for } k = 0, 1, 2, \ldots, n.$$

Observe also that the mean value theorem is the special case of Theorem 7.43 obtained by taking $n = 0$: on the interval $[c, x]$ there is a point z with

$$f(x) - f(c) = f'(z)(x - c).$$

Lagrange's result expresses the error term R_n in a particular way. It provides a sense of the error in approximating f by P_n. Note that we do not get an exact statement of the error term since it is given in terms of the value $f^{(n+1)}(z)$ at *some* point z. But if we know a little bit about the function $f^{(n+1)}$ on the interval in question, we might be able to say that this error is not very large.

Example 7.44: Suppose we wish to approximate the function $f(x) = \sin x$ on the interval $[-a, a]$ by a Taylor polynomial of degree 3, with $c = 0$. Here

$$f'(x) = \cos x\ ,\ f''(x) = -\sin x\ ,\ f'''(x) = -\cos x \text{ and } f^{(4)}(x) = \sin x.$$

Thus

$$\begin{aligned} P_3(x) &= \cos(0)x - \frac{\sin(0)}{2!}x^2 - \frac{\cos(0)}{3!}x^3 = x - \frac{x^3}{3!} \text{ and} \\ R_3(x) &= \frac{\sin z}{4!}x^4 \text{ for some } z \text{ in } [-a, a]. \end{aligned}$$

The exact error depends on which z makes this all true. But since $|\sin z| \leq 1$ for all z, we get immediately that

$$|R_3(x)| \leq a^4/4! = a^4/24,$$

so P_3 approximates f to within $a^4/24$ on the interval $[-a, a]$. For a small, the approximation should be sufficient for the purposes at hand. For large a, a higher-degree polynomial can produce the desired accuracy, since

$$|R_n(x)| \leq \frac{|x^{n+1}|}{(n+1)!}.$$

◀

Various other forms for the error term R_n are useful. The integral form is one of them. We state this form without proof. We assume that you are familiar with the integral as studied in calculus courses.

Theorem 7.45 (Integral Form of Remainder) *Suppose that the function f possesses at least $n+1$ derivatives on an open interval I and that $f^{(n+1)}$ is Riemann integrable on every closed interval contained in I. Let $c \in I$. Then*

$$R_n(x) = \frac{1}{n!}\int_c^x f^{(n+1)}(t)(x-t)^n\,dt \text{ for all } x \in I.$$

We shall see this form of the remainder again in Chapter 10 when we study Taylor series.

Exercises

7.12.1 Exhibit the Taylor polynomial about $x = 0$ of degree n for the function $f(x) = e^x$. Find n so that $|R_n(x)| \leq .0001$ for all $x \in [0,2]$.

7.12.2 Show that if f is a polynomial of degree n, then it is its own Taylor polynomial of degree n with $c = 0$.

7.12.3 Calculate the Taylor polynomial of degree 5 with $c = 1$ for the functions $f(x) = x^5$ and $g(x) = \ln x$.

7.12.4 Let $f(x) = \frac{1}{x+2}$, $c = -1$, and $n = 2$. Show that

$$\frac{1}{x+2} = 1 - (x+1) + (x+1)^2 + R_3$$

where, for some z between x and -1,

$$R_3 = -\frac{(x+1)^3}{(2+z)^4}.$$

7.12.5 Let $f(x) = \ln(1+x)$, $c = 0$, and $(x > -1)$. Show that

$$f(x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 + \cdots + (-1)^{n-1}\frac{x^n}{n} + R_n$$

where

$$R_n = \frac{(-1)^n}{n+1}\left(\frac{x}{1+z}\right)^{n+1}$$

for some z between 0 and x. Estimate R_n on the interval $[0, 1/10]$.

7.12.6 Just because a function possesses derivatives of all orders on an interval I does not guarantee that some Taylor polynomial approximates f in a neighborhood of some point of I. Let

$$f(x) = \begin{cases} e^{-\frac{1}{x^2}}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0. \end{cases}$$

(a) Show that f has derivatives of all orders and that $f^{(k)}(0) = 0$ for each $k = 0, 1, 2, \ldots$.

(b) Write down the polynomial P_n with $c = 0$.

(c) Write down Lagrange's form for the remainder of order n. Observe its magnitude and take the time to understand why P_n is not a good approximation for f on any interval I, no matter how large n is.

7.13 Challenging Problems for Chapter 7

7.13.1 **(Straddled derivatives)** Let $f : \mathbb{R} \to \mathbb{R}$ and let $x_0 \in \mathbb{R}$. Prove that f is differentiable at x_0 if and only if

$$\lim_{u \to x_0-,\, v \to x_0+} \frac{f(v) - f(u)}{v - u}$$

exists (finite), and, in this case, $f'(x_0)$ equals this limit.

7.13.2 **(Unstraddled Derivatives)** Let $f:\mathbb{R}\to\mathbb{R}$ and let $x_0\in\mathbb{R}$. We say f is strongly differentiable at x_0 if

$$\lim_{u\to x_0,\,v\to x_0,\,u\neq v}\frac{f(v)-f(u)}{v-u}$$

exists.

(a) Show that a differentiable function need not be strongly differentiable everywhere.

(b) Show that a strongly differentiable function must be differentiable.

(c) If f is strongly differentiable at a point x_0 and differentiable in a neighborhood of x_0, show that f' must be continuous there.

7.13.3 Let p be a polynomial of the nth degree that is everywhere nonnegative. Show that

$$p(x)+p'(x)+p''(x)+\cdots+p^{(n)}(x)\geq 0$$

for all x.

SEE NOTE 216

7.13.4 Suppose that f is continuous on $[0,1]$, differentiable on $(0,1)$, and $f(0)=0$ and $f(1)=1$. For every integer n show that there must exist n distinct points $\xi_1, \xi_2, \ldots, \xi_n$ in that interval so that

$$\sum_{k=1}^{n}\frac{1}{f'(\xi_k)}=n.$$

7.13.5 Show that there exists precisely one real number α with the property that for every function f differentiable on $[0,1]$ and satisfying $f(0)=0$ and $f(1)=1$ there exists a number ξ in $(0,1)$ (which depends, in general, on f) so that

$$f'(\xi)=\alpha\xi.$$

7.13.6 Let f be a continuous function. Show that the set of points where f is differentiable but not strongly differentiable (as defined in Exercise 7.13.2) is of the first category.

7.13.7 Let f be a continuous function on an open interval I. Show that f is convex on I if and only if

$$f\left(\frac{x+y}{2}\right)\leq\frac{f(x)+f(y)}{2}.$$

SEE NOTE 217

7.13.8 **(Wronskians)** The Wronskian of two differentiable functions f and g is the determinant

$$W(f,g)=\begin{vmatrix} f(x) & g(x)\\ f'(x) & g'(x)\end{vmatrix}.$$

Prove that if $W(f,g)$ does not vanish on an interval I and $f(x_1)=f(x_2)=0$ for points $x_1<x_2$ in I, then there exists $x_3\in(x_1,x_2)$ such that $g(x_3)=0$. [The functions $f(x)=\sin x, g(x)=\cos x$ furnish an example.]

SEE NOTE 218

7.13.9 ✂ Let f be a continuous function on an open interval I. Show that f is convex if and only if

$$\limsup_{h\to 0}\frac{f(x+h)+f(x-h)-2f(x)}{h^2}\geq 0$$

for every $x \in I$.

SEE NOTE 219

7.13.10 ✂ Let f be continuous on an interval (a,b).

(a) Prove that the four Dini derivates of f and the difference quotient $\frac{f(y)-f(x)}{y-x}$ $(x \neq y \in (a,b))$ have the same bounds.

(b) Prove that if one of the Dini derivates is continuous at a point x_0, then f is differentiable at x_0.

(c) Show by example that the statements in the first two parts can fail for discontinuous functions.

7.13.11 ✂ **(Denjoy-Young-Saks Theorem)** The theorem with this name is a far-reaching theorem relating the four Dini derivates D^+f, D_+f, D^-f and D_-f. It was proved independently by an English mathematician, Grace Chisolm Young (1868–1944), and a French mathematician, Arnaud Denjoy (1884–1974), for continuous functions in 1916 and 1915 respectively. Young then extended the result to a larger class of functions called measurable functions. Finally, the Polish mathematician Stanislaw Saks (1897–1942) proved the theorem for all real-valued functions in 1924. Here is their theorem.

> **Theorem (Denjoy-Young-Saks)** Let f be an arbitrary finite function defined on $[a,b]$. Then except for a set of measure zero every point $x \in [a,b]$ is in one of four sets:
>
> (1) A_1 on which f has a finite derivative.
>
> (2) A_2 on which $D^+f = D_-f$ (finite), $D^-f = \infty$ and $D_+f = -\infty$.
>
> (3) A_3 on which $D^-f = D_+f$ (finite), $D^+f = \infty$ and $D_-f = -\infty$.
>
> (4) A_4 on which $D^-f = D^+f = \infty$ and $D_-f = D_+f = -\infty$.

(a) Sketch a picture illustrating points in the sets A_2, A_3 and A_4. To which set does $x = 0$ belong when $f(x) = \sqrt{|x|}\sin x^{-1}$, $f(0) = 0$?

(b) Use the Denjoy-Young-Saks theorem to prove that an increasing function f has a finite derivative except on a set of measure zero.

(c) Use the Denjoy-Young-Saks theorem to show that if all derived numbers of f are finite except on a set of measure zero, then f is differentiable except on a set of measure zero.

(d) Use the Denjoy-Young-Saks theorem to show that, for every finite function f, the set $\{x : f'(x) = \infty\}$ has measure zero.

7.13.12 Let f be a continuous function on an interval $[a,b]$ with a second derivative at all points in (a,b). Let $a < x < b$. Show that there exists a point $\xi \in (a,b)$ so that

$$\frac{\frac{f(x)-f(a)}{x-a} - \frac{f(b)-f(a)}{b-a}}{x-b} = \tfrac{1}{2}f''(\xi).$$

SEE NOTE 220

7.13.13 Let $f : \mathbb{R} \to \mathbb{R}$ be a differentiable function with $f(0) = 0$ and suppose that $|f'(x)| \leq |f(x)|$ for all $x \in \mathbb{R}$. Show that f is identically zero.

SEE NOTE 221

7.13.14 Let $f:\mathbb{R}\to\mathbb{R}$ have a third derivative that exists at all points. Suppose that

$$\lim_{x\to\infty} f(x)$$

exists and that

$$\lim_{x\to\infty} f'''(x) = 0.$$

Show that

$$\lim_{x\to\infty} f'(x) = \lim_{x\to\infty} f''(x) = 0.$$

SEE NOTE 222

7.13.15 Let f be defined on an interval I of length at least 2 and suppose that f'' exists there. If $|f(x)|\le 1$ and $|f''(x)|\le 1$ for all $x\in I$ show that $|f'(x)|\le 2$ on the interval.
SEE NOTE 223

7.13.16 Let $f:\mathbb{R}\to\mathbb{R}$ be infinitely differentiable and suppose that

$$f\left(\frac{1}{n}\right) = \frac{n^2}{n^2+1}$$

for all $n = 1,2,3,\ldots$. Determine the values of

$$f'(0),\ f''(0),\ f'''(0),\ f^{(4)}(0),\ldots.$$

SEE NOTE 224

7.13.17 Let $f:\mathbb{R}\to\mathbb{R}$ have a third derivative that exists at all points. Show that there must exist at least one point ξ for which

$$f(\xi)f'(\xi)f''(\xi)f'''(\xi)\ge 0.$$

SEE NOTE 225

Notes

[161]Exercise 7.2.1. Write $x = x_0 + h$.

[162]Exercise 7.2.6. Write

$$\begin{aligned} &f(x+h)-f(x-h)\\ &= [f(x+h)-f(x)]+[f(x)-f(x-h)]. \end{aligned}$$

[163]Exercise 7.2.7. Use

$$1-\cos x = 2\sin^2 x/2.$$

When you take the square root be sure to use the absolute value.

[164]Exercise 7.2.12. Just use the definition of the derivative. Give a counterexample with $f(0)=0$ and $f'(0)>0$ but so that f is not increasing in any interval containing 0.

[165]Exercise 7.2.13. Even for polynomials, $p(x)$ increasing does not imply that $p'(x)>0$ for all x. For example, take $p(x)=x^3$. That has only one point where the derivative is not positive. Can you do any better?

[166]Exercise 7.2.14. Actually the assumptions are different. Here we assume $f'(x_0)$ does exist, whereas in the trapping principle we had to assume more inequalities to deduce that it exists.

[167]Exercise 7.2.15. Review Exercise 5.10.3 first.

[168]Exercise 7.2.16. Advanced (very advanced) methods would allow you to find a function continuous on $[0,1]$ that is differentiable at *no* point of that interval. For the purpose of this exercise just try to find one that is not differentiable at $1/2$, $1/3$, $1/4$, ... (Novices constructing examples often feel they need to give a simple formula for functions. Here, for example, you can define the function on $[1/2,1]$, then on $[1/4,1/2]$, then on $[1/8,1/4]$, and so on ... and then finally at 0.)

[169]Exercise 7.2.18. Find two examples of functions, one continuous and one discontinuous at 0, with an infinite derivative there.

[170]Exercise 7.2.19. Imitate the proof of Theorem 7.6. Find a counterexample to the question.

[171]Exercise 7.3.5. Use Theorem 7.7 (the product rule) and for the induction step consider

$$\frac{d}{dx}x^n = \frac{d}{dx}[x][x^{n-1}].$$

[172]Exercise 7.3.10. This formula is known as Leibniz's rule (which should indicate its age since Leibniz, one of the founders of the calculus, was born in 1646). It extends both Exercises 7.3.8 and 7.3.9. The formula is

$$(fg)^{(n)}(x_0)$$
$$= \sum_{k=0}^{n} \frac{n!}{k!(n-k)!} f^{(k)}(x_0)g^{(n-k)}(x_0).$$

[173]Exercise 7.3.11. Consider a sequence $x_n \to x_0$ with $x_n \neq x_0$ and $f(x_n) = f(x_0)$.

[174]Exercise 7.3.12. Let

$$f(x) = x^2 \sin x^{-1}$$

($f(0) = 0$) and take $x_0 = 0$. Utilize the fact that 0 is a limit point of the set $\{x : f(x) = 0\}$.

[175]Exercise 7.3.17. If $I(x)$ is the inverse function then $I(\sin x) = x$. The chain rule gives derivative as $I'(\sin x) = 1/\cos x$. This needs some work. Use

$$\cos x = \sqrt{1 - \sin^2 x}$$

and obtain

$$I'(\sin x) = \frac{1}{\sqrt{1 - \sin^2 x}}.$$

Now replace the $\sin x$ by some other variable. Caution: While doing this exercise make sure that you know how the arcsin function $\sin^{-1} x$ is actually defined. It is not the inverse of the function $\sin x$ since that function has no inverse.

[176]Exercise 7.3.19. Draw a good picture. The graph of $y = g(x)$ is the reflection in the line $y = x$ of the graph of $y = f(x)$. What is the slope of the reflected tangent line?

[177]Exercise 7.3.21. Use the idea in the example. If $f(x) = x^{1/m}$, then $[f(x)]^m = x$ and use the chain rule. If

$$F(x) = x^{n/m},$$

then

$$[F(x)]^m = x^n$$

and use the chain rule.

[178]Exercise 7.3.22. Once you know that

$$\frac{d}{dx}e^x = e^x$$

you can determine that

$$\frac{d}{dx}\ln x = 1/x$$

using inverse functions. Then consider $x^p = e^{p(\ln x)}$.

[179]Exercise 7.3.23. The formula you should obtain is

$$a_k = \frac{p^{(k)}(0)}{k!}$$

for $k = 0, 1, 2, \ldots$.

[180]Exercise 7.3.24. If you succeed, then you have proved the binomial theorem using derivatives. Of course, you need to compute $p(0)$, $p'(0)$, $p''(0)$, $p'''(0)$, ... to do this.

[181]Exercise 7.5.6. Define sets A_n consisting of all x for which $f(t) < f(x)$ for all $0 < |x-t| < \frac{1}{n}$ and observe that $\bigcup_{n=1}^{\infty} A_n$ is the set in question.

[182]Exercise 7.5.7. Modify the hint in Exercise 7.5.6.

[183]Exercise 7.6.3. Use Rolle's theorem to show that if x_1 and x_2 are distinct solutions of $p(x) = 0$, then between them is a solution of $p'(x) = 0$.

[184]Exercise 7.6.5. Use Rolle's theorem twice. See Exercise 7.6.7 for another variant on the same theme.

[185]Exercise 7.6.6. Since f is continuous we already know (look it up) that f maps $[a,b]$ to some closed bounded interval $[c,d]$. Use Rolle's theorem to show that there cannot be two values in $[a,b]$ mapping to the same point.

[186]Exercise 7.6.7. cf. Exercise 7.6.5.

[187]Exercise 7.6.8. First show directly from the definition that the Lipshitz condition will imply a bounded derivative. Then use the mean value theorem to get the converse, that is, apply the mean value theorem to f on the interval $[x,y]$ for any $a \le x < y \le b$.

[188]Exercise 7.6.9. Note that an increasing function f would allow only positive numbers in S.

[189]Exercise 7.6.12. Apply the mean value theorem to f on the interval $[x, x+a]$ to obtain a point ξ in $[x, x+a]$ with

$$f(x+a) - f(x) = af'(\xi).$$

[190]Exercise 7.6.13. Use the mean value theorem to compute

$$\lim_{x \to a+} \frac{f(x) - f(a)}{x - a}.$$

[191]Exercise 7.6.14. This is just a variant on Exercise 7.6.13. Show that under these assumptions f' is continuous at x_0.

[192]Exercise 7.6.15. Use the mean value theorem to relate

$$\sum_{i=1}^{\infty} (f(i+1) - f(i))$$

to

$$\sum_{i=1}^{\infty} f'(i).$$

Note that f is increasing and treat the former series as a telescoping series.

[193]Exercise 7.6.16. The proof of the mean value theorem was obtained by applying Rolle's theorem to the function

$$g(x) = f(x) - f(a) - \frac{f(b)-f(a)}{b-a}(x-a).$$

For this mean value theorem apply Rolle's theorem twice to a function of the form

$$h(x) = f(x) - f(a) - f'(a)(x-a) - \alpha(x-a)^2$$

for an appropriate number α.

[194]Exercise 7.6.18. Write

$$f(x+h) + f(x-h) - 2f(x) =$$
$$[f(x+h) - f(x)] + [f(x-h) - f(x)]$$

and apply the mean value theorem to each term.

[195]Exercise 7.6.21. Let $\phi(x)$ be

$$\begin{vmatrix} f(a) & g(a) & h(a) \\ f(b) & g(b) & h(b) \\ f(x) & g(x) & h(x) \end{vmatrix}$$

and imitate the proof of Theorem 7.21.

[196]Exercise 7.7.1. Interpret as a monotonicity statement about the function

$$f(x) = (1-x)e^x.$$

[197]Exercise 7.7.3. We do not assume differentiability at b. For example, this would apply to the function $f(x) = |x|$, with $b = 0$.

[198]Exercise 7.7.5. Interpret this as a monotonicity property for the function $F(x) = f(x)/x$. We need to show that F' is positive. Show that this is true if $f'(x) > f(x)/x$ for all x. But how can we show this? Apply the mean value theorem to f on the interval $[0,x]$ (and don't forget to use the hypothesis that f' is an increasing function).

[199]Exercise 7.7.6. If not, there is an interval $[a,b]$ with $f(a) = f(b) = 0$ and neither f nor g vanish on (a,b). Show that $f(x)/g(x)$ is monotone (increasing or decreasing) on $[a,b]$.

[200]Exercise 7.8.7. Let $\varepsilon > 0$ and consider $f(x) + \varepsilon x$.

[201]Exercise 7.8.9. Figure out a way to express $\mathbb{R}$ as a countable union of disjoint dense sets A_n and then let $f(x) = n$ for all $x \in A_n$. For an example subtract an appropriate linear function F from f such that $f - F$ is not an increasing function, and apply Theorem 7.30.

[202]Exercise 7.8.10. In connection with this exercise we should make this remark. If $A = \{a_k\}$ is any countable set, then the function defined by the series

$$\sum_{k=0}^{\infty} \frac{-|x-a_k|}{2^k}$$

has $D^+f(x) < D_-f(x)$ for all $x \in A$. This can be verified using the results in Chapter 9 on uniform convergence.

[203]Exercise 7.9.1. For the third part use the function $F(x) = x^2 \sin x^{-1}$, $F(0) = 0$ to show that there exists a differentiable function f such that $f'(x) = \cos x^{-1}$, $f(0) = 0$. Consider $g(x) = f(x) - x^3$ on an appropriate interval.

[204]Exercise 7.9.3. If either FG' or GF' were a derivative, so would the other be since

$$(FG)' = FG' + GF'.$$

In that case $FG' - GF'$ is also a derivative. But now show that this is impossible [because of (c)].

[205]Exercise 7.9.4. Use $fg' = (fg)' - f'g$. You need to know the fundamental theorem of calculus to continue.

[206]Exercise 7.9.5. If f' is continuous, then it is easy to check that E_α is closed. In the opposite direction suppose that every E_α is closed and f' is not continuous. Then show that there must be a number β and a sequence of points $\{x_n\}$ converging to a point z and yet $f'(x_n) \geq \beta$ and $f'(z) < \beta$. Apply the Darboux property of the derivative to show that this cannot happen if E_β is closed. Deduce that f' is continuous.

[207]Exercise 7.10.3. If f is convex on an interval I and g is convex *and also nondecreasing* on the interval $f(I)$, then you should be able to prove that $g \circ f$ is also convex. Show also that if the monotonicity assumption on g is dropped this might not be true.

[208]Exercise 7.10.5. Show that at every point of continuity of f'_+ the function is differentiable. How many discontinuities does the (nondecreasing) function f'_+ have?

[209]Exercise 7.10.10. Give an example of a convex function on the interval $(0,1)$ that is not bounded above; that answers the first question. For the second question use Exercise 7.10.4 to show that f must be bounded below.

[210]Exercise 7.10.13. The methods of Chapter 9 would help here. There we learn in general how to check for the differentiability of functions defined by series. For now just use the definitions and compute carefully.

[211]Exercise 7.10.14. For (d) let

$$f(x) = \begin{cases} e^{-1/x^2}(\sin 1/x)^2, & \text{for } x > 0 \\ 0, & \text{for } x = 0, \\ -e^{-1/x^2}(\sin 1/x)^2, & \text{for } x < 0 \end{cases} .$$

The three definitions in the exercise are not equivalent even for infinitely differentiable functions. They are, however, equivalent for *analytic* functions; that is, functions represented by power series (a topic we cover in Chapter 10). Since the scope of elementary calculus is more or less limited to functions that are analytic on the intervals on which the functions are concave up or down, we might argue that on that level, the definition to take is the one that is simplest to develop. We should mention, however, that there are differentiable functions that are not concave-up or concave-down on any interval!

[212]Exercise 7.10.15. Order the terms so that

$$x_1 \leq x_2 \leq \cdots \leq x_n.$$

And write

$$p = \sum_{k=1}^{n} \alpha_k x_k.$$

Choose a number M between $f'_-(p)$ and $f'_+(p)$. Check that

$$x_1 \leq p \leq x_n.$$

Check that

$$f(x_k) \geq M(x_k - p) + f(p)$$

for $k = 1, 2, \ldots, n$. Now use these inequalities to obtain Jensen's inequality.

[213]Exercise 7.11.1. Use L'Hôpital's rule to find that $f(0)$ should be $\ln(3/2)$. Use the definition of the derivative and L'Hôpital's rule twice to compute

$$f'(0) = [(\ln 3)^2 - (\ln 2)^2]/2.$$

Exercise 7.6.13 shows that the technique in (c) part does in fact compute the derivative provided only that you can show that this limit exists.

[214]Exercise 7.11.2. Treat the cases $A > 0$ and $A < 0$ separately.

[215]Exercise 7.11.10. We must have $\lim_{x \to \infty} f'(x) = 0$ in this case. (Why?)

[216]Exercise 7.13.3. Consider the function

$$H(x) = p(x) + p'(x) + p''(x) + \cdots + p^{(n)}(x)$$

and note, in particular, the relation between H, H' and p.

[217]Exercise 7.13.7. Such functions are called *midpoint convex*. By the definition of convexity we need to show that if $x_1, x_2 \in I$ and $\alpha \in [0, 1]$, then the inequality

$$f(\alpha x_1 + (1 - \alpha)x_2) \leq \alpha f(x_1) + (1 - \alpha) f(x_2)$$

is satisfied. Use the midpoint convexity condition to show that this is true whenever α is a fraction of the form $p/2^q$ for integers p and q. Now use continuity to show that it holds for all $\alpha \in [0, 1]$. Without continuity this argument fails and, indeed, there exist discontinuous midpoint convex functions that fail to be convex. [For an extensive account of what is known about such conditions, see B. S. Thomson, *Symmetric Properties of Real Functions*, Marcel Dekker, (New York, 1994).]

[218]Exercise 7.13.8. If g does not vanish on (x_1, x_2), then Rolle's theorem applied to the quotient f/g provides a contradiction. Incidentally, Josef de Wronski (1778–1853), whose name was attached firmly to this concept in 1882 in a multivolume *History of Determinants*, was a rather curious figure whom you are unlikely to encounter in any other context. One biographer writes about him:

> For many years Wronski's work was dismissed as rubbish. However, a closer examination of the work in more recent times shows that, although some is wrong and he has an incredibly high opinion of himself and his ideas, there is also some mathematical insights of great depth and brilliance hidden within the papers.

[219]Exercise 7.13.9. Consider the function

$$H(x) = f(x) + cx^2 + ax + b$$

for $c > 0$ and various choices of lines $y = ax + b$ and make use of Exercise 7.10.14.

[220]Exercise 7.13.12. This is from the 1939 Putnam Mathematical Competition.

[221]Exercise 7.13.13. This is from the 1946 Putnam Mathematical Competition.

[222]Exercise 7.13.14. This is from the 1958 Putnam Mathematical Competition.

[223]Exercise 7.13.15. This is from the 1962 Putnam Mathematical Competition.

[224]Exercise 7.13.16. This is from the 1992 Putnam Mathematical Competition.

[225]Exercise 7.13.17. This is from the 1998 Putnam Mathematical Competition.

Chapter 8

THE INTEGRAL

✂ For a short course the integral as conceived by Cauchy can be introduced and the material on Riemann's integral omitted or abridged. The study of the Riemann integral introduces new techniques and ideas that may not be needed for some courses.

8.1 Introduction

Calculus students learn two processes, both of which are described as "integration." The following two examples should be familiar:

$$\int x^3\,dx = x^4/4 + C$$

and

$$\int_1^2 x^3\,dx = 2^4/4 - 1^4/4 = 16/4 - 1/4 = 15/4.$$

The first is called an *indefinite integral* or *antiderivative* and the second a *definite integral*. The use of nearly identical notation, terminology, and methods of computation does a lot to confuse the underlying meanings. Many calculus students would be hard pressed to make a distinction.

Indeed even for many eighteenth-century mathematicians these two different procedures were not much distinguished. It was a great discovery that the computation of an area could be achieved by finding an antiderivative. It is attributed to Newton, but vague ideas along this line can be found in the thinking of earlier authors. For these mathematicians a definite integral was defined directly in terms of the antiderivative.

This is most unfortunate for the development of a rigorous theory, as recognized by Cauchy. He saw clearly that it was vital that the meaning of the definite integral be separated from the indefinite integral and given a precise definition independent of it. For this he turned to the geometry of the Greeks, who had long ago described a method for computing areas of regions enclosed by curves. This method, the so-called method of exhaustion, involves computing the areas of simpler figures (squares, triangles, rectangles) that approximate the area of the region.

We return to the example

$$\int_1^2 x^3\,dx$$

interpreted as an area. The region is that bounded on the left and right by the lines $x = 1$ and $x = 2$, below by the line $y = 0$, and above by the curve $y = x^3$. (See Figure 8.1.)

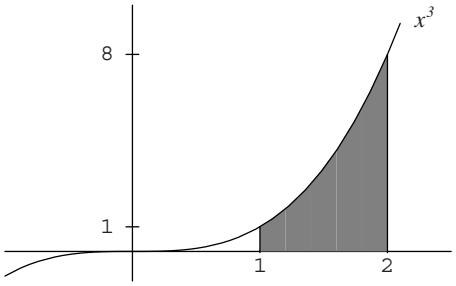


Figure 8.1. Region bounded by $x = 1$, $x = 2$, $y = x^3$, and $y = 0$.

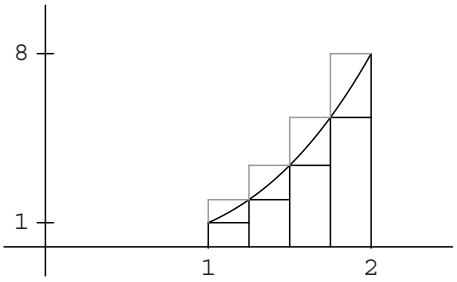


Figure 8.2. Method of exhaustion ($n = 4$).

Using the method of exhaustion, we may place this figure inside a collection of rectangles by dividing the interval $[1,2]$ into n equal sized subintervals each of length $1/n$. This means selecting the points

$$1,\ 1+1/n,\ 1+2/n,\ \ldots,\ 1+(n-1)/n$$

and constructing rectangles with the height of the rectangles determined by right hand endpoints. The total area of these rectangles exceeds the true area and is precisely

$$\sum_{k=1}^{n}(1+k/n)^3(1/n).$$

The method of exhaustion requires a lower estimate as well and the true area of the region must be greater than

$$\sum_{k=1}^{n}(1+(k-1)/n)^3(1/n).$$

(See Figure 8.2 for an illustration with $n = 4$.)

The method of exhaustion requires us to show that as n increases *both* approximations, the upper one and the lower one, approach the same number. Cauchy saw that, because of the continuity of the function $f(x) = x^3$, these limits would be the same. More than that, any choice of points ξ_k from the interval $[1+(k-1)/n, 1+(k)/n]$

would have the property that

$$\lim_{n\to\infty} \sum_{k=1}^{n} {\xi_k}^3 (1/n)$$

would exist.

This procedure, borrowed heavily from the Greeks, will work for any continuous function and thus it offered to Cauchy a way to *define* the integral

$$\int_a^b f(x)\,dx$$

for any function f, continuous on an interval $[a,b]$, without any reference whatsoever to notions of derivatives or antiderivatives. The key ingredients here are first dividing the interval $[a,b]$ by a finite sequence of points

$$a = x_0 < x_1 < x_2 < x_3 < \cdots < x_{n-1} < x_n = b,$$

thus forming a collection of nonoverlapping subintervals called *a partition* of $[a,b]$

$$[x_0,x_1],\ [x_1,x_2],\ \ldots,\ [x_{n-1},x_n],$$

(it is not important that they have equal size, just that they get small). Then we form the sums

$$\sum_{k=1}^{n} f(\xi_k)(x_k - x_{k-1}) \tag{1}$$

with respect to this partition. The only constraint on the choice of the points ξ_k is that each is taken from the appropriate interval $[x_{k-1},x_k]$ of the partition; these are often called the *associated points*. It is an unfortunate trick of fate that the sums (1) that originated with Cauchy are called *Riemann sums* because of Riemann's later (much later) use of them in defining his integral.

In this chapter we start with Cauchy's methods of integration and proceed to Riemann. The important thing for you to keep track of is how this theory develops in a manner that assigns meaning to the integral of various classes of functions in a way distinct from how we would compute an integral in a calculus course. It is much easier to compute that $\int_1^2 x^3\,dx = 15/4$ in the familiar way, rather than as a limit of Riemann sums; but the meaning of this statement is properly given in this more difficult way.

8.2 Cauchy's First Method

Cauchy's first goal in defining an integral was to give meaning to the integral for continuous functions. The integral is defined as the limit of Riemann sums. Before such a definition is valid we must show that the limit exists. Thus the first step is the proof of the following theorem.

Theorem 8.1 (Cauchy) *Let f be a continuous function on an interval $[a,b]$. Then there is a number I, called the* definite integral *of f on $[a,b]$, such that for each $\varepsilon > 0$ there is a $\delta > 0$ so that*

$$\left| \sum_{k=1}^{n} f(\xi_k)(x_k - x_{k-1}) - I \right| < \varepsilon$$

whenever $[x_0,x_1]$, $[x_1,x_2]$, $\ldots$, $[x_{n-1},x_n]$ *is a partition of the interval* $[a,b]$ *into subintervals of length less than* δ *and each* ξ_k *is a point in the interval* $[x_{k-1},x_k]$.

Once the theorem is proved, then we can safely define the definite integral of a continuous function as that number I guaranteed by the theorem. Loosely speaking, we say that the integral is defined as a limit of Riemann sums (1).

Definition 8.2: Let f be a continuous function on an interval $[a,b]$. Then we define

$$\int_a^b f(x)\,dx$$

to be that number I whose existence is proved in Theorem 8.1.

Now we must prove Theorem 8.1.

Proof. For any particular partition (let us call it π)

$$[x_0,x_1],\ [x_1,x_2],\ \ldots,\ [x_{n-1},x_n]$$

of the interval $[a,b]$ write

$$M(\pi) = \sum_{k=1}^{n} \max\{f(x) : x \in [x_{k-1},x_k]\}(x_k - x_{k-1})$$

and

$$m(\pi) = \sum_{k=1}^{n} \min\{f(x) : x \in [x_{k-1},x_k]\}(x_k - x_{k-1}).$$

Here $M(\pi)$ and $m(\pi)$ depend on the partition π. These are called the *upper sums* and *lower sums* for the partition. Note that any Riemann sum over this partition must lie somewhere between the lower sum and the upper sum.

Since f is continuous on $[a,b]$ it is uniformly continuous there (Theorem 5.48). Thus for every $\varepsilon > 0$ there is a $\delta > 0$ that depends on ε so that

$$|f(x) - f(y)| < \frac{\varepsilon}{b-a}$$

if $|x-y| < \delta$. Since we shall need to find a different δ for many choices of ε, let us write it as $\delta(\varepsilon)$.

Thus if the partition we are using has the property that every interval is shorter than $\delta(\varepsilon)$, we must have

$$\max\{f(x) : x \in [x_{k-1},x_k]\} - \min\{f(x) : x \in [x_{k-1},x_k]\} < \frac{\varepsilon}{b-a}$$

It follows that for such partitions $0 \le M(\pi) - m(\pi) < \varepsilon$.

Select a sequence of partitions $\{\pi_n\}$, each one containing all the points of the previous partition, and such that every interval in the nth partition π_n is shorter than $\delta(1/n)$. If $M(\pi_n)$ and $m(\pi_n)$ denote the corresponding sums for the nth partition of our sequence of partitions, then

$$0 \le M(\pi_n) - m(\pi_n) < 1/n.$$

One more technical point needs to be raised. As we add points to a partition the upper sums cannot increase nor can the lower sums decrease. Thus $M(\pi_n) \ge M(\pi_{n+1})$